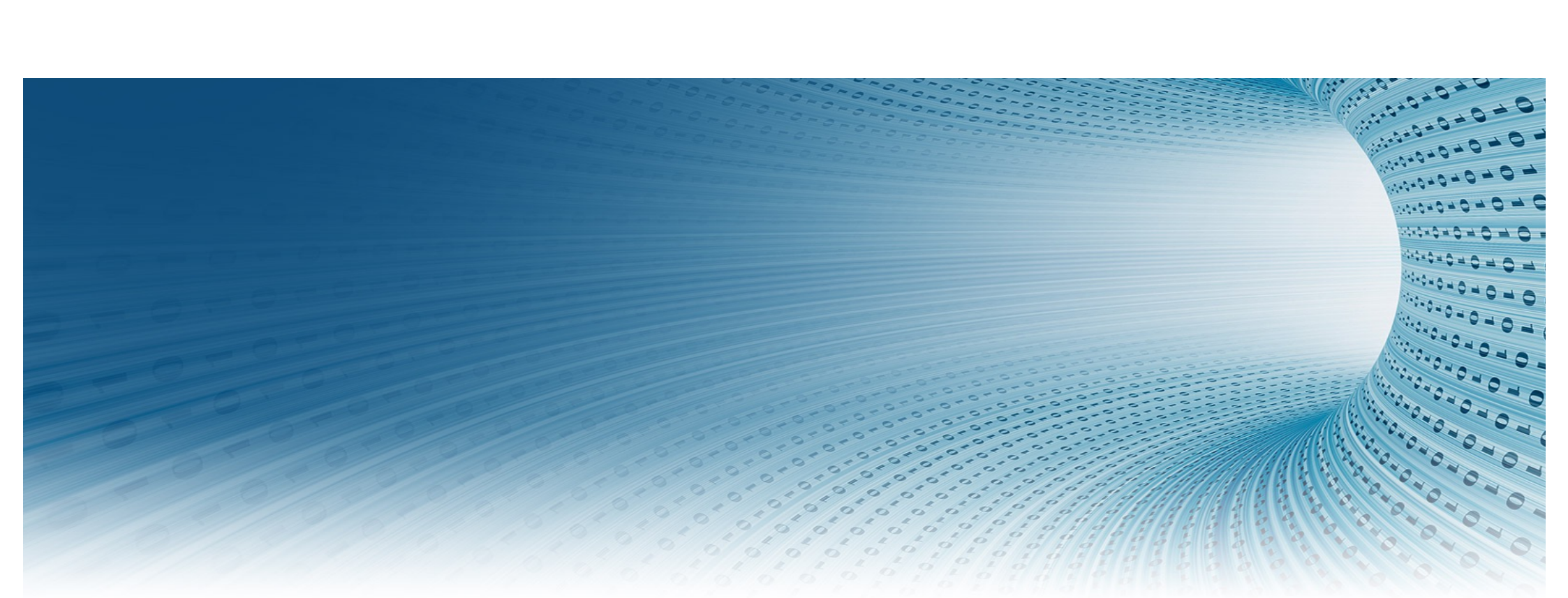




Trajectories of Markov Chains

Theory and formulation

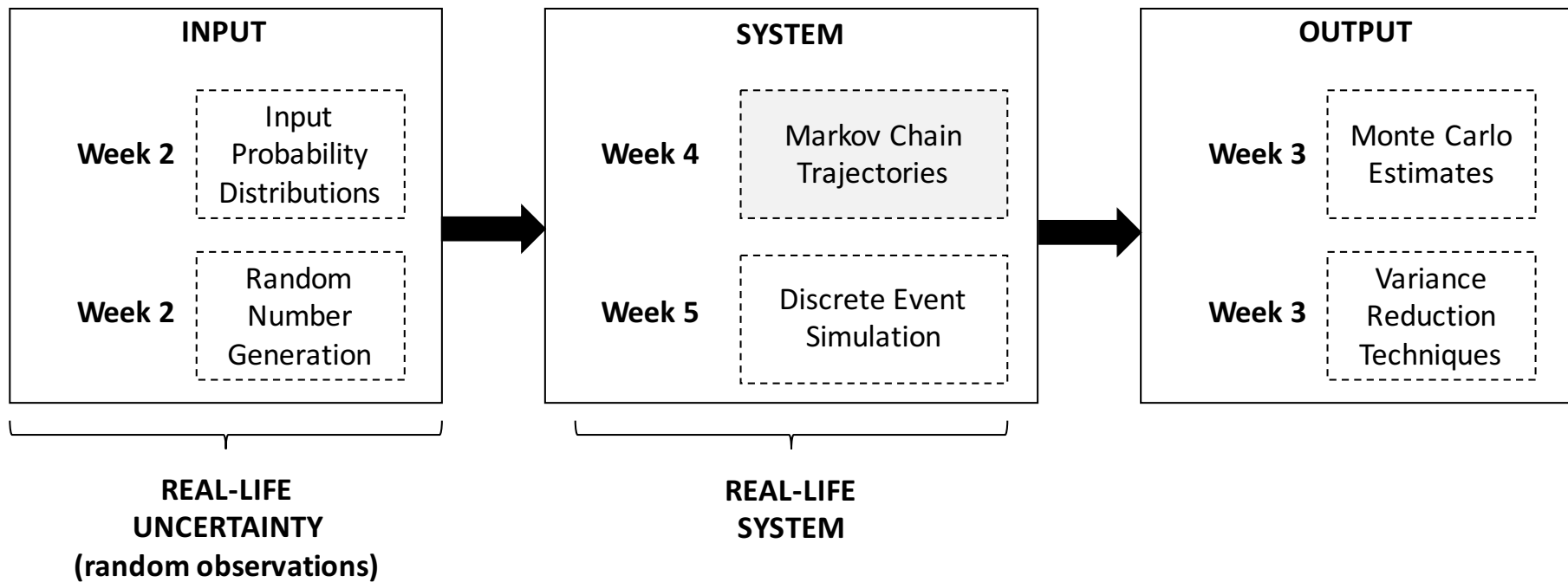
Broos Maenhout

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Tweekerkenstraat 2

9000 Gent

Outline

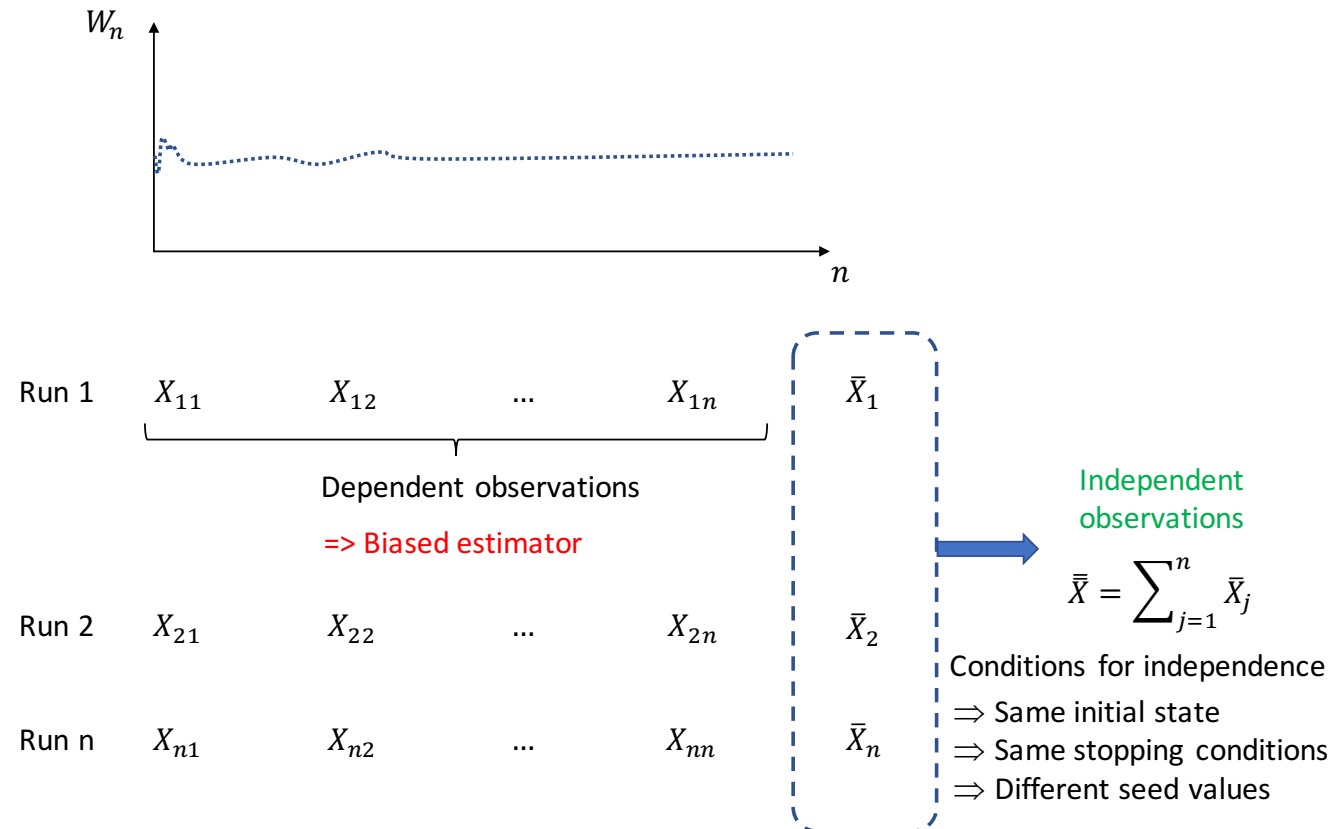


Sampling

- Dependence between observations

- Example

(not in your slides)



Outline

- Definition
- Finite state space Markov Chains
- Countable state space Markov Chains
- Ergodic chains
- Estimators for Markov Chains
- Variance reduction for Markov Chains

Definition of Markov Chains

- Markov Chains involve probability models for stochastic processes, i.e. processes that evolve over time in a probabilistic manner
- A stochastic process is a sequence of dependent random variables $\{Y_k \in \Omega\}$

with

K	the set of time periods (index k) -	Discrete time stochastic process ($k \in \mathbb{Z}^+$) vs Continuous time stochastic process ($k \in \mathbb{R}^+$)
Y_k	a measurable characteristic at time k -	E.g. market demand, currency exchange rate, monthly inventory level
Ω	the set of states of the process -	The state space can be finite or infinite

Definition of Markov Chains

- Example

(not in your slides)

- A machine is either working or in repair. If it is working at time k , there is a chance β that it will be working at time $k + 1$. If it is being repaired, there is a chance $1 - \alpha$ that it will work at time $k + 1$.

- Let

$Y_k = 1$ when the machine is working at time k and

$Y_k = 0$ when it is being repaired

- One-step probability matrix

$$\begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} \alpha & 1-\alpha \\ 1-\beta & \beta \end{bmatrix} \end{matrix}$$

- Assume $\alpha = 0.1$ and $\beta = 0.8$

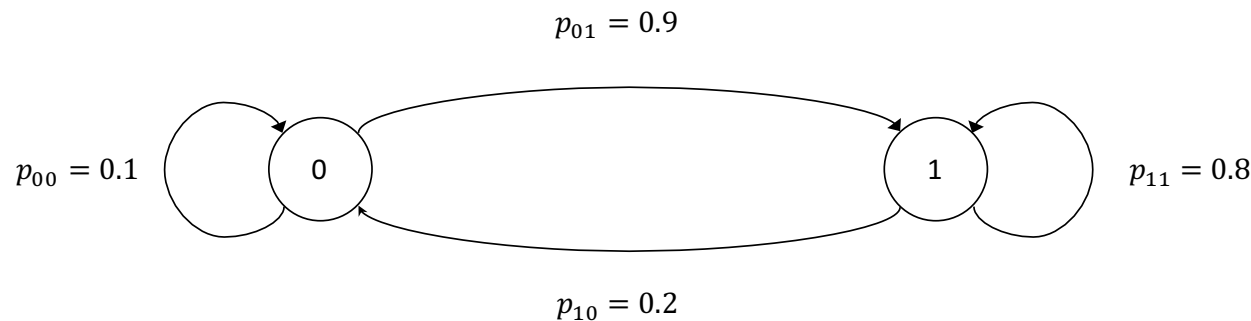
$$\begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

Definition of Markov Chains

- Example

(not in your slides)

- A machine is either working or in repair. If it is working at time k , there is a chance β that it will be working at time $k + 1$. If it is being repaired, there is a chance $1 - \alpha$ that it will work at time $k + 1$.
- State transition diagram



Definition of Markov Chains

- Dependent random variable

- General stochastic process

- The evolution over time of a (discrete time) stochastic process is described by the (one-step) transition probabilities

$$P(Y_{k+1} = j \mid Y_k = i, Y_{k-1} = s_{k-1}, \dots, Y_0 = s_0)$$

- The process will move from a state to other states with known transition probabilities.
- If these probabilities do not change over time, then the transition probabilities are stationary

- Markovian Property

- If the next state Y_{k+1} of the process depends only on the present state Y_k and is independent of past history, the process is called a Markov Chain
- A stochastic process has the *Markovian property* if the transition probabilities depend on the current state only

$$P(Y_{k+1} = j \mid Y_k = i, Y_{k-1} = s_{k-1}, \dots, Y_0 = s_0) = P(Y_{k+1} = j \mid Y_k = i)$$

Definition of Markov Chains

- Dependent random variable

- Markov Chains

- A Markov Chain is a stochastic process
- A Markov Chain $\{Y_k \in \Omega\}$ is a sequence of random variables, recursively defined by

$$Y_{k+1} = \phi(Y_k, X_k) \quad \text{with } Y_0 = s_0$$

where $\{X_k \in \Theta\}$ is a sequence of independent and identically distributed random variables

$\phi: \Omega \times \Theta \rightarrow \Omega$ is a deterministic function

s_0 is an independent random variable

- Note: Let $\phi_n = \phi(y, X_n)$ to simplify notation

Finite state space Markov Chains

- A Markov Chain is completely defined by (assume $\Omega = \{\mathbf{1}, \mathbf{2}, \dots, K\}$)

➤ A set of transition probabilities

$$\begin{aligned} p_{ij} &= \Pr[Y_{k+1} = j \mid Y_k = i] \\ &= \Pr[\phi(Y_k, X_k) = j \mid Y_k = i] \\ &= \Pr[\phi(i, X_k) = j] \end{aligned}$$

The matrix $P = [p_{ij}]_{i,j=1,\dots,K}$ is the one-step transition matrix

$$P = \begin{bmatrix} p_{11} & \dots & p_{1K} \\ & \dots & \\ p_{K1} & \dots & p_{KK} \end{bmatrix}$$

➤ The initial probability vector

$$s^0 = [s_1^0, \dots, s_K^0] = [\Pr[Y_0 = i]]_{i=1,\dots,K}$$

Finite state space Markov Chains

- A Markov Chain is completely defined by (assume $\Omega = \{1, 2, \dots, K\}$)

➤ A set of transition probabilities

➤ The initial probability vector

➤ Example

(not in your slides)

$$s_1 = s_0 P \quad \begin{bmatrix} 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.9 \end{bmatrix}$$

Case 1

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.2 & 0.8 \end{bmatrix}$$

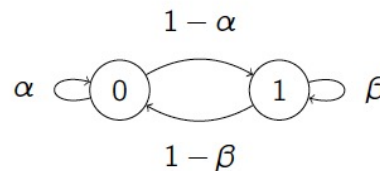
Case 2

$$\begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.15 & 0.85 \end{bmatrix}$$

Case 3

Finite state space Markov Chains

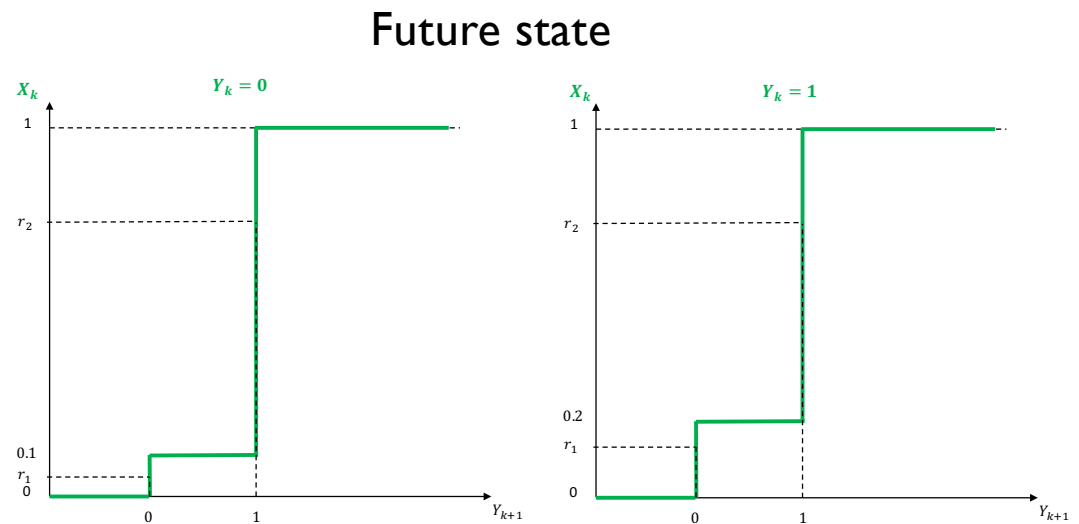
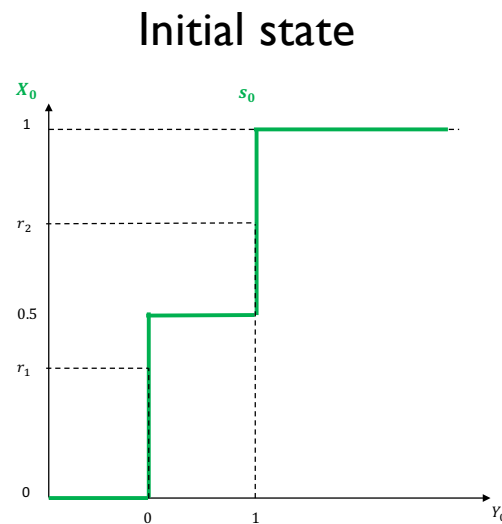
- Assume that a Markov Chain with transition matrix P and initial probability vector s^0 is given. An observation is generated as follows
 - Let X_k be a sequence of uniformly distributed random variables
 - Y_k is generated as follows
 - Set $Y_0 = j$ if $\sum_{\ell=1}^{j-1} s_{\ell}^0 \leq X_0 < \sum_{\ell=1}^j s_{\ell}^0$
 - Set $Y_{k+1} = j$ if $\sum_{\ell=1}^{j-1} p_{Y_k \ell} \leq X_{k+1} < \sum_{\ell=1}^j p_{Y_k \ell}$
- The state transition diagram
 - A graphical representation for discrete-time Markov Chains



Finite state space Markov Chains

- Assume that a Markov Chain with transition matrix P and initial probability vector s^0 is given. An observation is generated as follows:

➤ Example



Infinite state space Markov Chains

- Infinite state space: Example – The Galton-Watson process

- Branching stochastic process to model the extinction of family names

- Assumptions

- Family names are passed from father to son
- Offspring are randomly either male or female
 - The number of a man's sons is a random variable $O_{k,\ell} = \{0, 1, 2, 3, \dots\}$ following a Poisson distribution with parameter λ
 - The number of different men's sons are independent random variables all having the same distribution
- Names become extinct if the family name dies out

- Model

$$Y_{k+1} = \sum_{\ell=1}^{Y_k} O_{k,\ell} = \phi_k(Y_k)$$

with Y_k the number of individuals in the k^{th} generation

$O_{k,\ell}$ the number of offspring of the ℓ^{th} individual in the k^{th} generation

State space is
infinite but
countable

Infinite state space Markov Chains

- Infinite state space: Example – The Galton-Watson process

➤ Input

- We assume that the offspring is Poisson distributed with parameter $\lambda = 1.2$
- # generations = 100
- $s_0 = 5$

➤ Pseudocode

Initialise parameters ()

```
Procedure(){  
  for k in range (1,K)  
    printf(k, y, sum/k);  
    y = phi(y);  
    sum += y;}
```

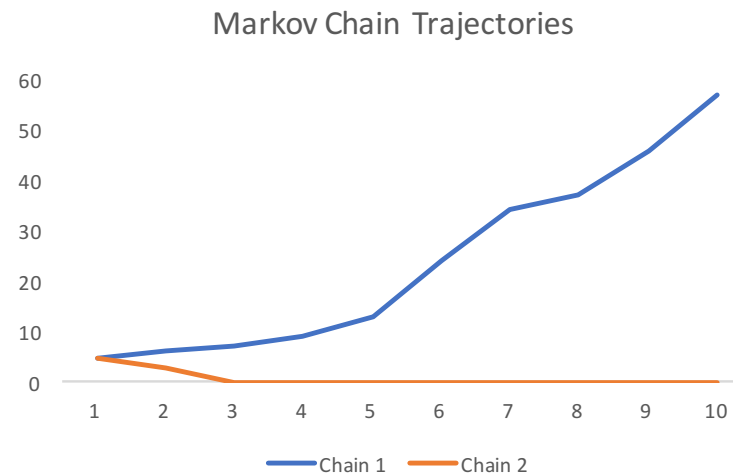
```
phi(y){  
  offspring = 0  
  for i in range (1, y)  
    offspring += Poisson_distribution(lambda);  
  return offspring;}
```

Infinite state space Markov Chains

- Infinite state space: Example – The Galton-Watson process

➤ Results: Sample path of a branching process ($K = 100$, $s_0 = 5$, $\lambda = 1.2$)

- Trajectory 1: Red line
- Trajectory 2: Blue line



- It can be shown that the population either grows to infinity or dies out
- A single trajectory is not enough to study such a process

Infinite state space Markov Chains

- Infinite state space: Example – Waiting times in a queue

➤ Lindley equation: The consecutive waiting times relate as

$$Y_{k+1} = \max(Y_k + S_k - A_k, 0)$$

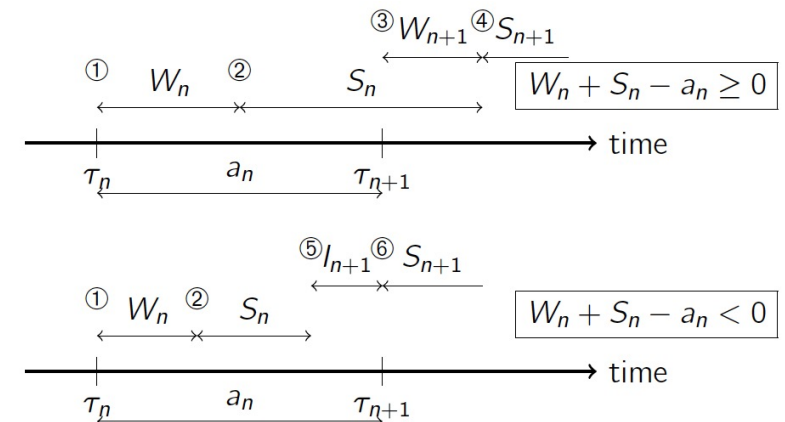
State space is
infinite

with

Y_k the waiting time of the k^{th} customer that arrives at a queue

S_k the service time of the k^{th} customer in the queue

A_k the time between the arrival instant of the k^{th} customer and the $k+1^{\text{st}}$ customer



Infinite state space Markov Chains

- Infinite state space: Example – Waiting times in a queue

- Interarrival times: Gamma distribution – $\mu_A = 10$ (mean); $\sigma_A^2 = 20$ (variance)

- Service times: Gamma distribution – $\mu_S = 8$ (mean); $\sigma_S^2 = 50$ (variance)

- Gamma distribution

- Probability density function

$$f_X(x) = f(x; \beta, \alpha) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} \quad \text{for } x > 0$$

with α the shape parameter

β the scale parameter

- To estimate the parameters of the gamma distribution that best fits the sampled data, the following formulae can be used

$$\alpha = \mu^2 / \sigma^2$$

$$\beta = \sigma^2 / \mu$$

Infinite state space Markov Chains

- Infinite state space: Example – Waiting times in a queue

➤ Pseudocode

```
Initialise parameters (){
```

```
    sum = y = 0;
```

```
     $\mu_A = 10$ ;  $\sigma_A^2 = 8$ ;
```

```
     $\mu_S = 10$ ;  $\sigma_S^2 = 50$ ;
```

```
     $\alpha_S = \mu_S^2 / \sigma_S^2$ ;  $\beta_S = \sigma_S^2 / \mu_S$ ;
```

```
     $\alpha_A = \mu_A^2 / \sigma_A^2$ ;  $\beta_A = \sigma_A^2 / \mu_A$ ;
```

```
    N = 2000;
```

```
}
```

```
Procedure(){
```

```
    for k in range (1,K)
```

```
        printf(k, y, sum/k);
```

```
        y = phi(y);
```

```
        sum += y;}
```

```
phi(y){
```

```
    return max (y + gamma( $\alpha_S$ ,  $\beta_S$ ) – gamma( $\alpha_A$ ,  $\beta_A$ ), 0;}
```

Infinite state space Markov Chains

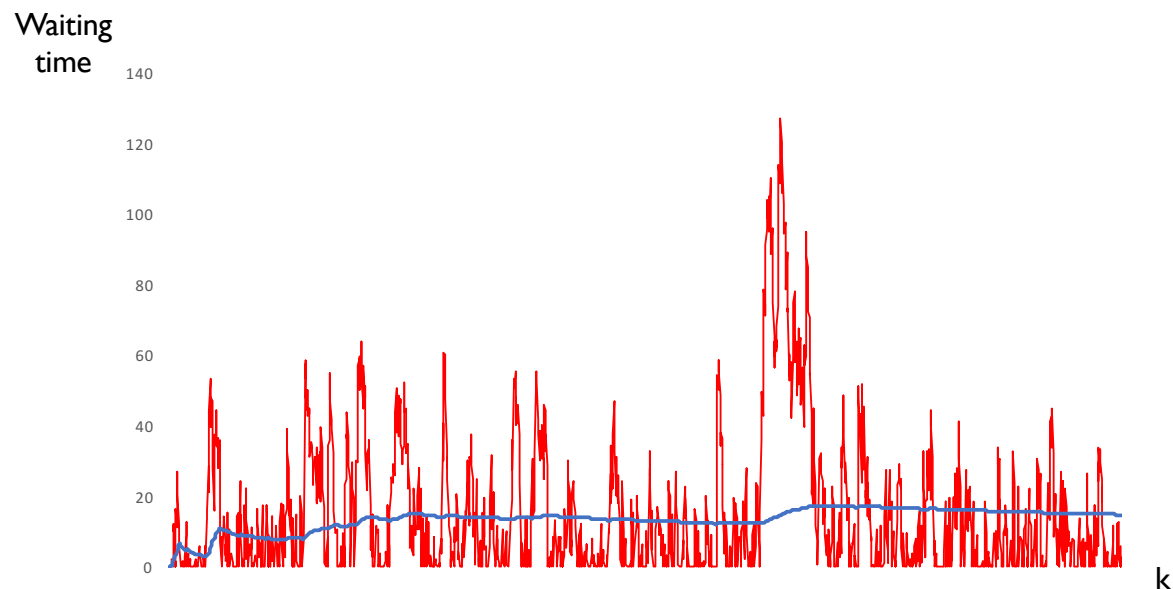
- Infinite state space: Example – Waiting times in a queue

- Probability distributions

- Interarrival times: Gamma distribution – $\mu_A = 10$ (mean); $\sigma_A^2 = 20$ (variance)
- Service times: Gamma distribution – $\mu_S = 8$ (mean); $\sigma_S^2 = 50$ (variance)

- Results

- Markov trajectory: Red line
- Running average: Blue line



Infinite state space Markov Chains

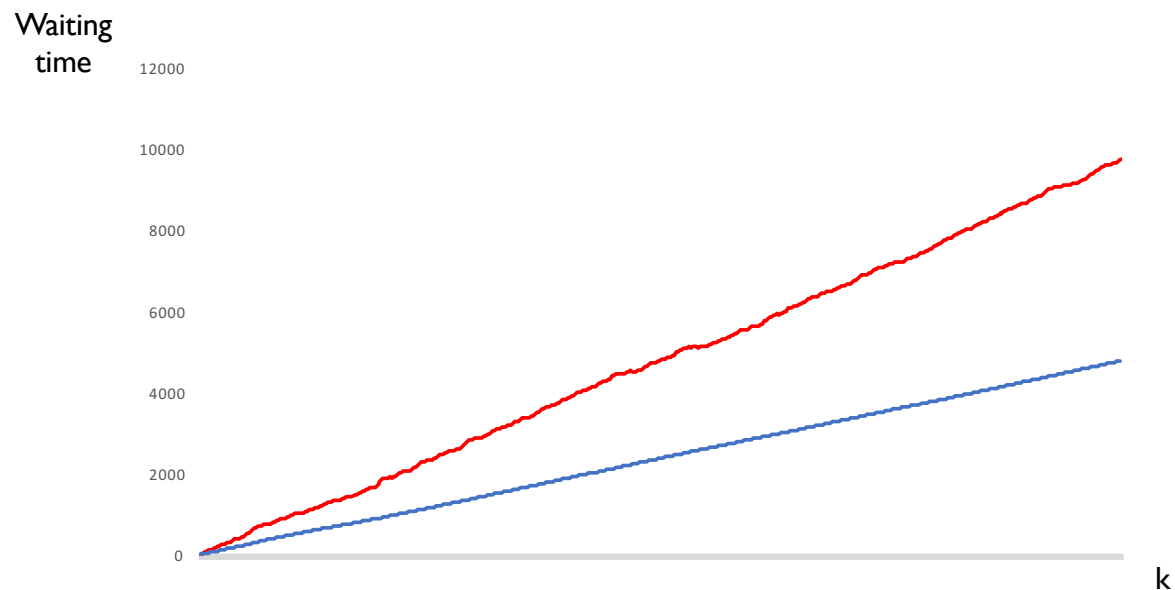
- Infinite state space: Example – Waiting times in a queue

- Probability distributions

- Interarrival times: Gamma distribution – $\mu_A = 10$ (mean); $\sigma_A^2 = 20$ (variance)
- Service times: Gamma distribution – $\mu_S = 15$ (mean); $\sigma_S^2 = 50$ (variance)

- Results

- Markov trajectory: Red line
- Running average: Blue line



Countable state space Markov Chains

- n-step transition probabilities

- Probability of arriving in state j after n steps starting from state i: $\mathbf{p}_{ij}^{(n)} = P(Y_n = j \mid Y_0 = i)$

- Chapman-Kolmogorov equations

$$p_{ij}^{(n)} = \sum_{k=1}^N p_{ik}^{(m)} p_{kj}^{(n-m)}$$

$$p_{ij}^{(n)} = \sum_{k=1}^N p_{ik} p_{kj}^{(n-1)} = \sum_{n=1}^N p_{ik}^{(n-1)} p_{kj}$$

- The n-step probabilities can be obtained recursively from the one-step transition probabilities (with P the transition matrix).

$$P^{(2)} = P \times P$$

$$P^{(3)} = P \times P \times P$$

- The n-step transition probability matrix $P(n)$ is given by the n' th power of P:

$$P^{(n)} = P^n = P \times P^{(n-1)} = P^{(n-1)} \times P =$$

$$\text{or } P^{(n)} = P^{(m-k)} \times P^{(k)} \quad \text{where } n = m + k$$

$$P^{(n)} = \begin{pmatrix} p_{00}^{(n)} & p_{01}^{(n)} \\ p_{10}^{(n)} & p_{11}^{(n)} \end{pmatrix}$$

- If the one-step probabilities are stationary, the n-step transition probabilities are also stationary:

$$\mathbf{p}_{ij}^{(n)} = P(X_{t+n} = j \mid X_t = i) = P(X_n = j \mid X_0 = i)$$

Countable state space Markov Chains

- Example

(not in your slides)

- Machine-repair example

- One-step probability matrix = P

$$\begin{matrix} & & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

- Two-step probability matrix = $P \times P$

$$\begin{matrix} & & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.19 & 0.81 \\ 0.18 & 0.82 \end{bmatrix} \end{matrix}$$

- Five step probability matrix

$$\begin{matrix} & & 0 & 1 \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 2/11 & 9/11 \\ 2/11 & 9/11 \end{bmatrix} \end{matrix} =$$

Independent from
initial state
=> Steady-state
probabilities

Countable state space Markov Chains

➤ Exercise 2 (see exercise bundle MCT)

The following matrix is the single-step transition probability matrix of a discrete-time Markov Chain which describes the weather. State 1 represents a sunny day, state 2 a cloudy day and state 3 a rainy day.

- One-step probability matrix

	Sunny	Cloudy	Rainy
Sunny	0.7	0.2	0.1
Cloudy	0.3	0.5	0.2
Rainy	0.2	0.6	0.2

- Two-step probability matrix

$$P^{(2)} = P \times P$$

$$\begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.6 & 0.2 \end{bmatrix} \times \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.6 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.57 & 0.3 & 0.13 \\ 0.4 & 0.43 & 0.17 \\ 0.36 & 0.46 & 0.18 \end{bmatrix}$$

- Ten-step probability matrix

$$P^{(10)} = P^{(9)} \times P$$

	0	1	2
0	28/59	22/59	9/59
1	28/59	22/59	9/59
2	28/59	22/59	9/59

Countable state space Markov Chains

- Classification of states

- Classes of communication

- State j is accessible from state i (denoted as $i \rightarrow j$) if it can be reached from i , i.e. $p_{ij}^{(n)} > 0$ for some n
- States i and j are said to communicate if $i \rightarrow j$ and $j \rightarrow i$. $\exists n: p_{ij}^{(n)} > 0$
- States can be partitioned into communicating classes such that communicating states are in the same class.
- A Markov chain is irreducible if all states communicate, i.e. if there is only one class.

- Machine repair example (not in your slides)

1. Any state communicates with itself
2. If state i communicates with state j , then state j communicates with state i
3. If state i communicate with state j and state j communicates with state k , then state i communicates with state k .

Countable state space Markov Chains

- Classification of states

- Periodicity

- The period of state i is the integer t ($t > 1$) such that $p_{ii}^{(n)} = 0$ for all values of n other than $t, 2t, 3t, \dots$ ($t = \gcd\{n, p_{ii}^{(n)} > 0\}$)
 - If there are two consecutive numbers s and $s+1$ such that the process can be in state i at times s and $s+1$, the state is called aperiodic (i.e. the state has period 1).

n	1	2	3	4		(not in your slides)
$p_{ii}^{(n)}$	0	>0	0	>0	...	(periodic system)
$p_{ii}^{(n)}$	>0	>0	>0	>0	...	(aperiodic system)

- Periodicity is a class property, i.e. all states in a class either have the same period or are all aperiodic.
 - A Markov Chain on a finite state space is ergodic if it is irreducible and aperiodic.

Countable state space Markov Chains

- Example

(not in your slides)

- Machine-repair example

One-step probability matrix = P

$$\begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

Two-step probability matrix

$$\begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 0.19 & 0.81 \\ 0.18 & 0.82 \end{bmatrix} \end{matrix}$$

n	1	2	
$p_{00}^{(n)}$	0.1	0.19	(aperiodic state)
$p_{11}^{(n)}$	0.8	0.82	(aperiodic state)

Countable state space Markov Chains

- Recurrent states

- Let f_{ij}^n be the probability to travel to state j at time n , starting from state i at time 0 without visiting state j before time n

$$f_{ij}^n = \Pr[\min\{k \geq 1: X_k = j\} = n | X_0 = i]$$

- These probabilities relate to the transition probabilities as

$$f_{ij}^n = \sum_{k, k \neq j} p_{ik} f_{kj}^{n-1} \quad \text{for } n = 2, 3, \dots$$

$$f_{ij}^1 = p_{ij}$$

$$\sum_{n=1}^{\infty} f_{ij}^n = f_{ij} = p_{ij} + \sum_{k, k \neq j} p_{ik} f_{kj} \quad \text{with } f_{ij} \quad \text{the passage or hitting probability } i \rightarrow j$$

- A state is recurrent if one returns with probability one, i.e.

$$\sum_{n=1}^{\infty} f_{ii}^n = f_{ii} = 1$$

Otherwise, the state is transient ($f_{ii} < 1$)

Countable state space Markov Chains

- Weather example

(not in your slides)

	Sunny	Cloudy	Rainy
Sunny	0.7	0.2	0.1
Cloudy	0.3	0.5	0.2
Rainy	0.2	0.6	0.2

- Calculate hitting probabilities

$$f_{ij} = p_{ij} + \sum_{k \neq j} p_{ik} f_{kj}$$

$$f_{ii} = ? \quad (\text{probability that we return to the same state})$$

e.g. $i = 1$

$$f_{11} = 0.7 + 0.2f_{21} + 0.1f_{31}$$

$$f_{21} = 0.3 + 0.5f_{21} + 0.2f_{31}$$

$$f_{31} = 0.2 + 0.6f_{21} + 0.2f_{31}$$

$$\Rightarrow f_{11} = 1, f_{21} = 1, f_{31} = 1$$

$\Rightarrow$ Same exercise to calculate f_{22} and f_{33}



Countable state space Markov Chains

- Recurrent states

- The mean recurrence time of a recurrent state i is defined by the mean time between consecutive visits to state i , i.e.

$$E[T_i] = E[\min\{n > 0: Y_n = i \text{ given } Y_0 = i\}]$$

$$E[T_i] = \sum_{n=1}^{\infty} n f_{ii}^n$$

$$E[T_i] = \mu_{ii} = 1 + \sum_{k, k \neq i} p_{ik} \mu_{ki} \quad \text{with } \mu_{ii} \text{ first passage time } i \rightarrow i$$

- Properties

- Mean recurrence time is finite for irreducible Markov Chains on a finite state space
- If the mean recurrence time is finite ($\Pr[T_i < \infty] = 1$ and $E[T_i] < \infty$), the state is positive recurrent
- If a state is positive recurrent, the expected number of visits to the state is infinite
- A Markov Chain is transient, positive recurrent or null recurrent if the chain is irreducible and its states are transient, positive recurrent or null recurrent respectively

Countable state space Markov Chains

- Weather example

(not in your slides)

	Sunny	Cloudy	Rainy
Sunny	0.7	0.2	0.1
Cloudy	0.3	0.5	0.2
Rainy	0.2	0.6	0.2

- Calculate expected number of steps before we go to the same state again (= first passage times)

$$\mu_{ij} = 1 + \sum_{k \neq j} p_{ik} \mu_{kj}$$

$$\mu_{ii} = ? \quad (\text{recurrence time state } i)$$

e.g. $i = 1$

$$\mu_{11} = 1 + 0.2\mu_{21} + 0.1\mu_{31}$$

$$\mu_{21} = 1 + 0.5\mu_{21} + 0.2\mu_{31}$$

$$\mu_{31} = 1 + 0.6\mu_{21} + 0.2\mu_{31} \quad \Rightarrow \mu_{11} = 59/28, \mu_{21} = 25/7, \mu_{31} = 55/14$$

$$\text{and further} \quad \Rightarrow \mu_{12} = 59/22, \mu_{22} = 45/11, \mu_{32} = 25/11$$

$$\Rightarrow \mu_{13} = 59/9, \mu_{23} = 70/9, \mu_{33} = 20/3$$

$\Rightarrow$ All states are positive recurrent since $\Pr[T_i < \infty] = 1$ and $E[T_i] < \infty$



Countable state space Markov Chains

- A Markov Chain on a finite state space is ergodic if
 - There is only one communicating class (irreducible)
 - Aperiodicity
 - All states are positive recurrent
- ⇒ Ergodicity is key for the existence of the limiting distribution

Countable state space Markov Chains

(not in your slides)

Ergodicity:

Markov
Chain with a
FINITE
state space

1. All states are communicating (Irreducibility)

$$p_{ij}^{(n)} > 0 \text{ and } p_{ji}^{(n)} > 0 \quad \exists n, \forall (i, j)$$

2. Aperiodicity

n	1	2	3	4	5
-----	---	---	---	---	---

$p_{ii}^{(n)}$	0	> 0	0	> 0	0
----------------	---	-----	---	-----	---

(Period = 2)

All probabilities are equal to 0,
other than 2, 4, 6, 8, etc

$p_{ii}^{(n)}$	...	> 0	> 0	...	...
----------------	-----	-----	-----	-----	-----

(aperiodic)

Two consecutive
probabilities > 0

Example:

0.7	0.2	0.1
0.3	0.5	0.2
0.2	0.6	0.2

Countable state space Markov Chains

(not in your slides)

Ergodicity:

Markov Chain
with a **INFINITE**
state space

- 1+2. (Irreducibility + Aperiodicity)
- 3. Positive recurrence

$$f_{ii} = 1 \quad \forall i$$

$$= \Pr[T_i < \infty] = 1$$

(hitting probability $\Rightarrow$ we will for sure visit a recurrent state again)

$$E[T_i] < \infty$$

POSITIVE RECURRENT

$$\lambda < \mu$$

Infinite state
space

$\leftarrow W_n$

W_{n+1}

Limiting distribution
exists

$$E[T_i] = +\infty$$

NULL RECURRENT

$$\lambda \geq \mu$$

Infinite state
space

$\leftarrow W_n$

W_{n+1}

Explosive MC
more arrivals than
can be processed

No limiting distribution

Ergodic Chains

- Limiting distribution

➤ Consider the limits p_{ij}^n for $n \rightarrow \infty, \forall i, j$

$$\pi_j = \lim_{n \rightarrow \infty} p_{ij}^n$$

➤ For ergodic Markov Chains

- The n-step probabilities converge when n increases, such that all the rows of the n-step probability matrix are identical.

$$\pi = \lim_{n \rightarrow \infty} p^n \quad \text{with } p^n = [\Pr[Y_n = i]]_{i=1, \dots, K} = s^0 P^n$$

- i.e. for large enough n, the probability that the system is in state j no longer depends on the initial state s^0 at time 0. These probabilities are the steady-state probabilities and they cumulate to 1.

Ergodic Chains

- For Ergodic Markov Chains

- The limiting distribution is a stationary distribution for the Markov Chain with transition probabilities p_{ij} if

$$\pi_j = \sum_i \pi_i p_{ij} \quad \text{and} \quad \sum_j \pi_j = 1$$

- i.e. the steady-state probabilities do not change from transition to transition and they cumulate to 1. This gives a set of $K+1$ equations for K unknowns, so one equation is redundant. The limiting distribution is thus the unique normalised solution of $\pi = \pi P$
 - Y_n and Y_{n+1} have the same distribution, i.e. since one step later the distribution remains the same we call the distribution stationary.
- For a function f , we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{n=1}^N f(Y_n) = E[f(Y)]$$

with $Y \sim \pi$ i.e. average over the sample path yields the steady state mean value

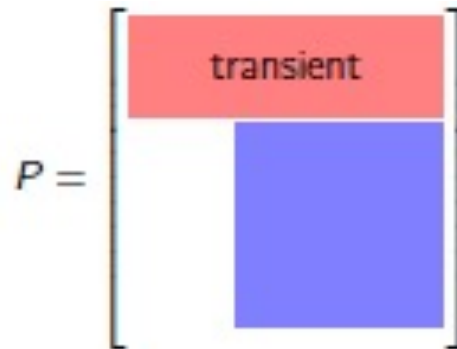
Ergodic Chains

- For Ergodic Markov Chains

➤ Recurrence times

$$E[T_i] = 1/\pi_i$$

➤ Illustration



Ergodic Chains

- What if the chain is not ergodic?
 - Classification of states if the chain is not irreducible
 - Transient $\Pr[T_i < \infty] < 1$
 - Positive recurrent $\Pr[T_i < \infty] = 1$ and $E[T_i] < \infty$
 - Null recurrent $\Pr[T_i < \infty] = 1$ and $E[T_i] = \infty$
 - Illustration

$$P = \begin{bmatrix} \text{transient} & & \\ & \text{blue square} & \\ & & \text{blue square} \end{bmatrix}$$

Estimator for Markov Chains

- Across-replication data versus within-replication data

- Within-replication data

Y_{ij} jth observation in the ith replication

- Across-replication data are formed by summarising within-replication data

$\bar{Y}_i$ average of the ith replication

Y_i^{max} maximum value in the ith replication over all observations

- Across-replication data are independent (different random numbers) and identically distributed (same model),

- In Markov Chains, within-replication data are not independent and identically distributed

(E.g. cycle time for parts and work in process (WIP), waiting time)

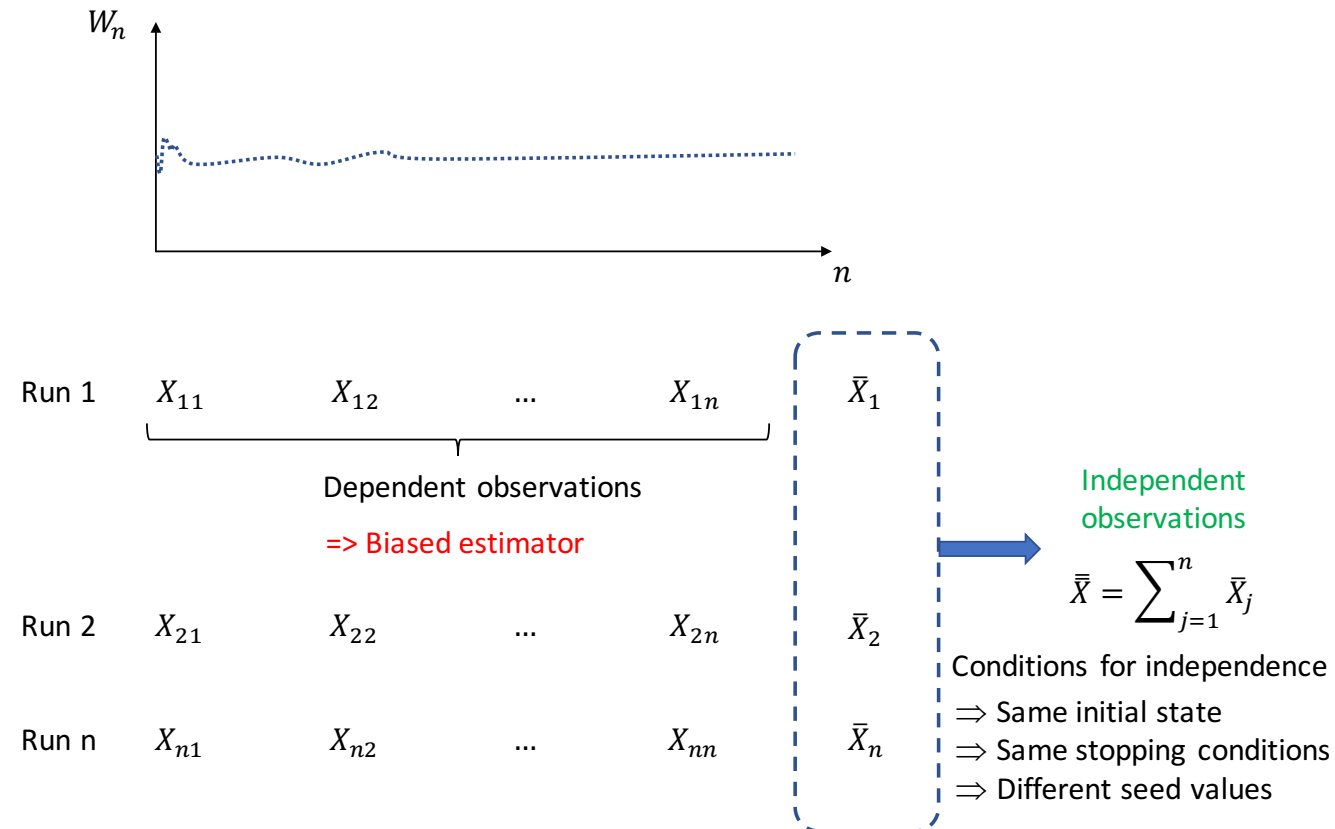
=> Multiple Markov trajectories need to be evaluated

Sampling

- Dependence between observations

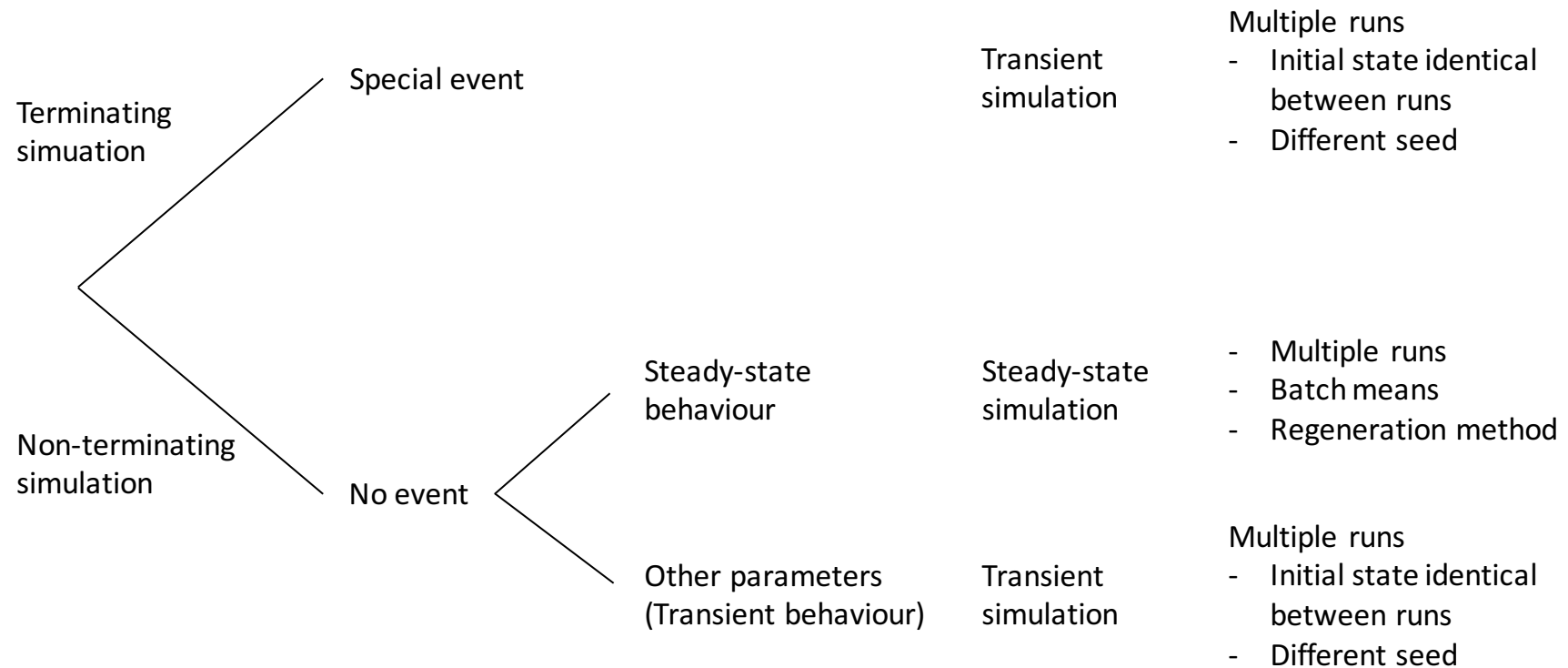
- Example

(not in your slides)



Estimator for Markov Chains

(not in your slides)



Estimator for Markov Chains

- How to evaluate steady-state behaviour?

- The estimator $\bar{X}$ is a consistent estimate of the steady-state behaviour for ergodic Markov Chains

$$\bar{X} = \frac{1}{N} \sum_{n=1}^N f(Y_n) \text{ and } E[\bar{X}] = \frac{1}{N} \sum_{n=1}^N f(Y_n) = \mu$$

The sample mean of the stationary process converges to the mean of the limiting distribution π

- Methods

- Replication method: Multiple runs
- Batch means method: 1 run => More efficient and more accurate methods
- Regenerative method: 1 run => More efficient and more accurate methods

Estimator for Markov Chains

- Transient period

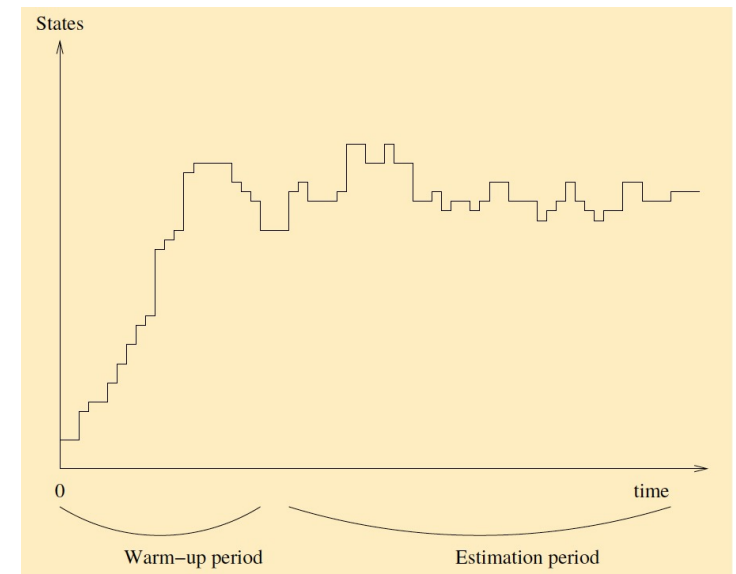
- The estimator $\bar{X}$ is biased

$$\pi_n \neq \pi \quad \text{and} \quad \bar{X} = E[f(Y_n)] \neq E[f(Y)] = \mu$$

- For n sufficiently large $\pi_n \approx \pi$

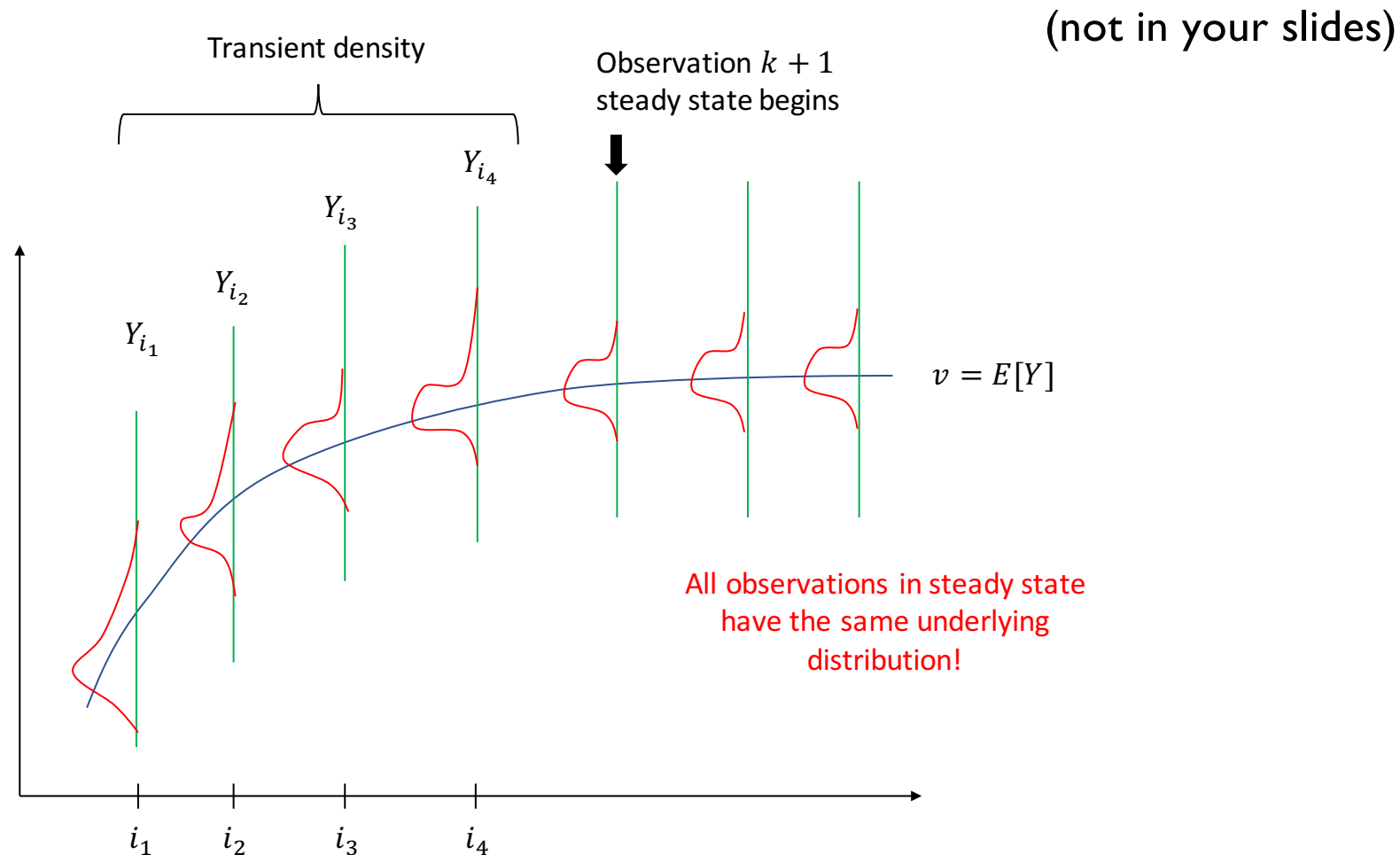
$$E[\bar{X}] = \lim_{n \rightarrow N} \frac{1}{N} \sum_{n=N_1+1}^{N_1+N} f(Y_n) = \mu$$

- Discard the first N_1 samples until the chain is sufficiently close to the limiting distribution
- Discard the transient behaviour or the warm-up period



Estimator for Markov Chains

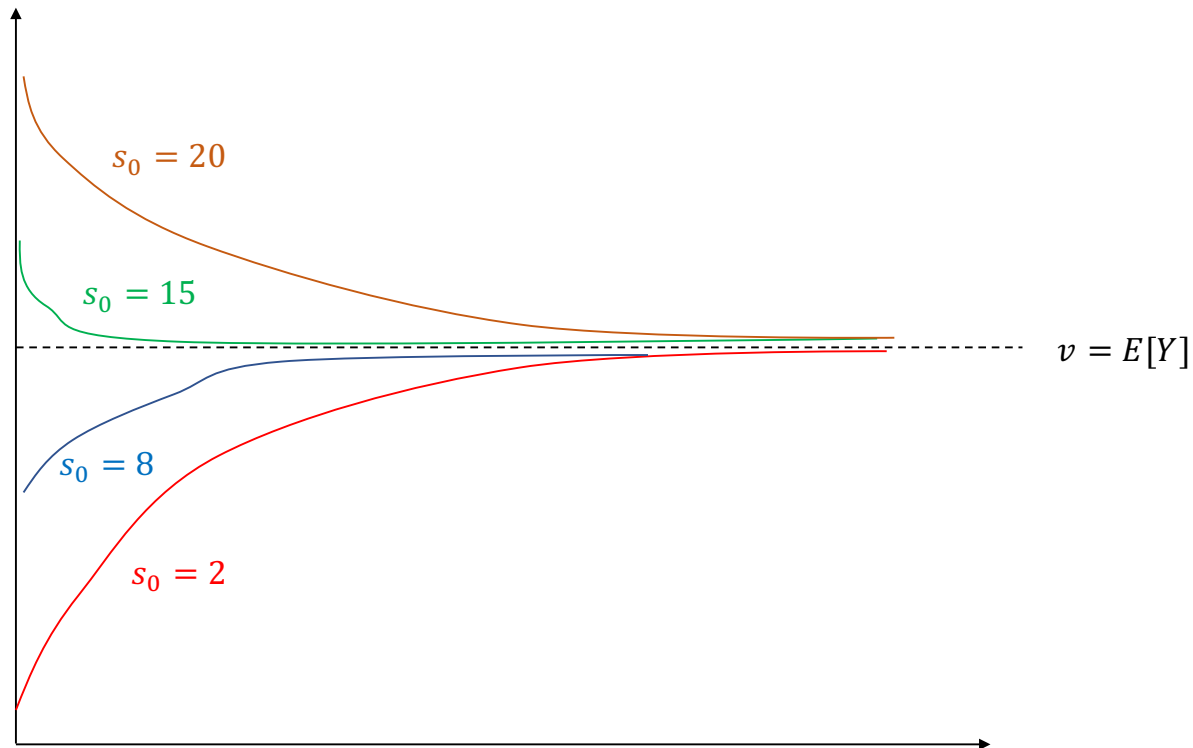
- Transient period



Estimator for Markov Chains

- Transient period

(not in your slides)



Estimator for Markov Chains

- Warm-up period
 - In theory, the transient period lasts infinitely long. But after a reasonable time (N_t samples), the difference between the distribution Y_k and the stationary distribution becomes negligible
 - When are we close to the limiting distribution? Too long vs too short
 - The influence of the initial distribution has disappeared
 - Not trivial: Based on the eigenvalue (cf. Grassmann (2011)) or the simpler algorithm of Welch (1981) may be used
 - It is possible to generate steady-state samples via perfect sampling
 - Warm-up period makes it time consuming to get multiple independent samples of the limiting distribution
- ⇒ Better to generate only one sample and split it into several batches
- ⇒ Account for the dependence between the consecutive samples

References:

Grassmann, W.K., 2011, Rethinking the Initialisation Bias Problem in Steady-State Discrete Event Simulation, Proc. 2011 Winter Simulation Conference, Phoenix, Arizona, 593-599

Welch, P.D., 1981, On the problem of the Initial Transient in Steady-State Simulation, IBM Watson Research Center, Yorktown Heights, New York

Estimator for Markov Chains

- Warm-up period
 - Graphical methods: warm up length is decided by examining the time-series output of statistics of interest
 - Heuristic approaches: use simple rules to determine warm up length
 - Statistical methods
 - Initialisation bias tests: check whether the early data in the time series is biasing the calculation of statistics of interest
 - Hybrid methods: combinations of an initialisation bias test and another truncation method to decide on warm up length

References:

Robinson, S. 2004. *Simulation: The Practice of Model Development and Use*. John Wiley and Sons Ltd.

Hoad, K., S. Robinson, and R. Davies. 2010. "Automating Warm-Up Length Estimation". *Journal of the Operational Research Society* 61:1389–1403.

Estimator for Markov Chains

- Warm-up period

- Algorithm of Welch (1981)

- The method uses the moving average of the output statistic and looks for the point t_0 in the time series at which the moving average flattens out (i.e. Warm-up period = $[0, t_0]$)
 - The warm-up period is determined based upon the average of multiple runs ($n \geq 5$) due to variability
 - The length T of one simulation run is much larger than the anticipated length of the warm-up period t_0
 - Assume

Y_{it} observation on time point t of simulation run i

$\bar{Y}_t$ average observation on time point t over multiple simulation runs, i.e. $\bar{Y}_t = \frac{1}{n} \sum_{i=1}^n Y_{it}$

Estimator for Markov Chains

- Warm-up period

- Algorithm of Welch (1981)

- Step 1. Run the simulation model n times, where $n \geq 5$ **to obtain time-series of the output** Y , i.e. Y_{it}
- Step 2. Calculate the mean of the output data in each period, i.e. $\bar{Y}_t = \frac{1}{n} \sum_{i=1}^n Y_{it}$
- Step 3. Calculate a moving average for the $\bar{Y}_t$ using a window size w , where $w \leq \lfloor T/4 \rfloor$ (Law 2007)
- Step 4. Plot the moving average $\bar{Y}_t(w)$ as a time series

$$\bar{Y}_t(w) = \begin{cases} \sum_{\tau=1}^{2t-1} \frac{\bar{Y}_\tau}{2t-1} & t = 1, \dots, w \\ \sum_{\tau=0}^{2w} \frac{\bar{Y}_{t-w+\tau}}{2w+1} & t = w+1, \dots, T-w \end{cases}$$

- Step 5. Analyse the plot, if the plot is not smooth, increase the window size w and repeat steps 3 and 4
- Step 6. The warm up period t_0 is the point at which the time series of the moving average becomes flat

Estimator for Markov Chains

- Warm-up period

- Algorithm of Welch (1981)

(not in your slides)

- Step 4. Plot the moving average $\bar{Y}_t(w)$ as a time series

$$\bar{Y}_t(w) = \begin{cases} \sum_{\tau=1}^{2t-1} \frac{\bar{Y}_\tau}{2t-1} & t = 1, \dots, w \\ \sum_{\tau=0}^{2w} \frac{\bar{Y}_{t-w+\tau}}{2w+1} & t = w+1, \dots, T-w \end{cases}$$

- E.g. $w = 100$ (window size)

if $t > 100$ # periods = $2w + 1 = 201$

=> $t = 250$ average over periods 150 – 350

if $t \leq 100$ # periods = $2t - 1$

=> $t = 2$ average over periods 1, 2 and 3

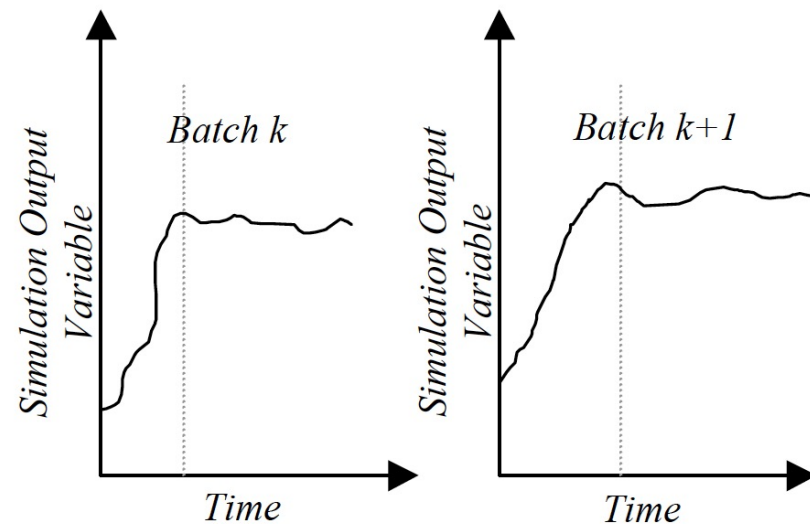
=> $t = 3$ average over periods 1, 2, 3, 4 and 5

Estimator for Markov Chains

- Estimation Method: Replication method

- Principle

- $N (> 25)$ independent runs of the simulation experiment
 - Remove transient period from each run and generate estimators for steady-state behaviour

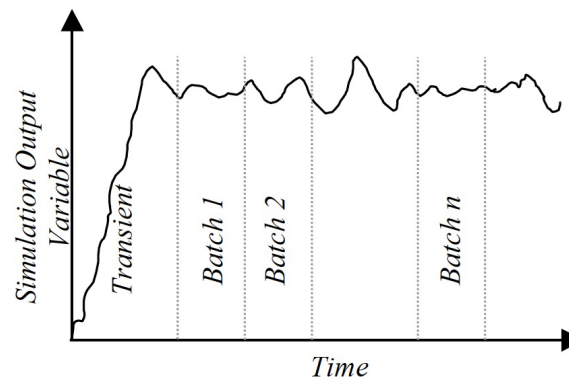


Estimator for Markov Chains

- Estimation Method: Batch mean method

- Principle

- 1 single run of the Markov Chain/Discrete Event Simulation Model
 - The sequence of N samples $\{Y_1, Y_2, \dots, Y_N\}$ is partitioned into L batches of M samples ($N = M \times L$)



- Estimators

- The sample mean taken over the ℓ th batch equals

$$\bar{X}_l = \frac{1}{M} \sum_{k=(l-1)M+1}^{lM} f(Y_k)$$

Estimator for Markov Chains

- Estimation Method: Batch mean method

➤ Principle

Dependent

(not in your slides)

$X_{i-2}, X_{i-1}, X_i, X_{i+1}, X_{i+2}$

=> neighbouring observations strongly correlated!

=> $|\text{Cov}(X_{i-2}, X_i)|$ high

Independent

X_i, X_{i+100}

=> observation far away from each other

=> $|\text{Cov}(X_i, X_{i+100})|$ low close to 0

$\Rightarrow |\text{Cov}(X_{i-2}, X_i)| \gg |\text{Cov}(X_i, X_{i+100})|$

How long one batch? Observations should be **independent**

Find L such that $|\text{Cov}(X_i, X_{i+L})| \approx 0$

Length one batch = 5L

Estimator for Markov Chains

- Estimation Method: Batch mean method

- Estimators

- The mean of the batch sample means is equal to the sample mean of the complete sequence

$$\bar{X} = \frac{1}{N} \sum_{n=1}^N f(Y_k) = \frac{1}{L} \sum_{l=1}^L \frac{1}{M} \sum_{k=(l-1)M+1}^{lM} f(Y_k) = \frac{1}{L} \sum_{l=1}^L \bar{X}_l$$

- The variance of the sample mean can be estimated by

$$\text{var}[\bar{X}] = \frac{1}{L-1} \sum_{l=1}^L (\bar{X}_l - \bar{X})^2$$

(We assume that M is sufficiently large such that the sequence of batch means constitute a sequence of almost independent random variables)

- Confidence interval

$$\Pr \left[\bar{X} - k_p \sqrt{\text{var}[\bar{X}]} \leq \mu \leq \bar{X} + k_p \sqrt{\text{var}[\bar{X}]} \right] = p$$

i.e. the likelihood that the error is larger than k_p times the standard deviation is p

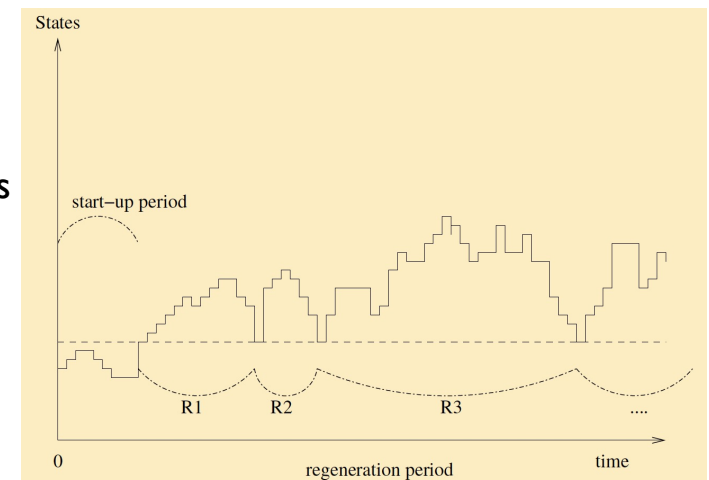
(k_p is found in the tables of the normal distribution)

Estimator for Markov Chains

- Estimation Method: Regeneration method

- Principle

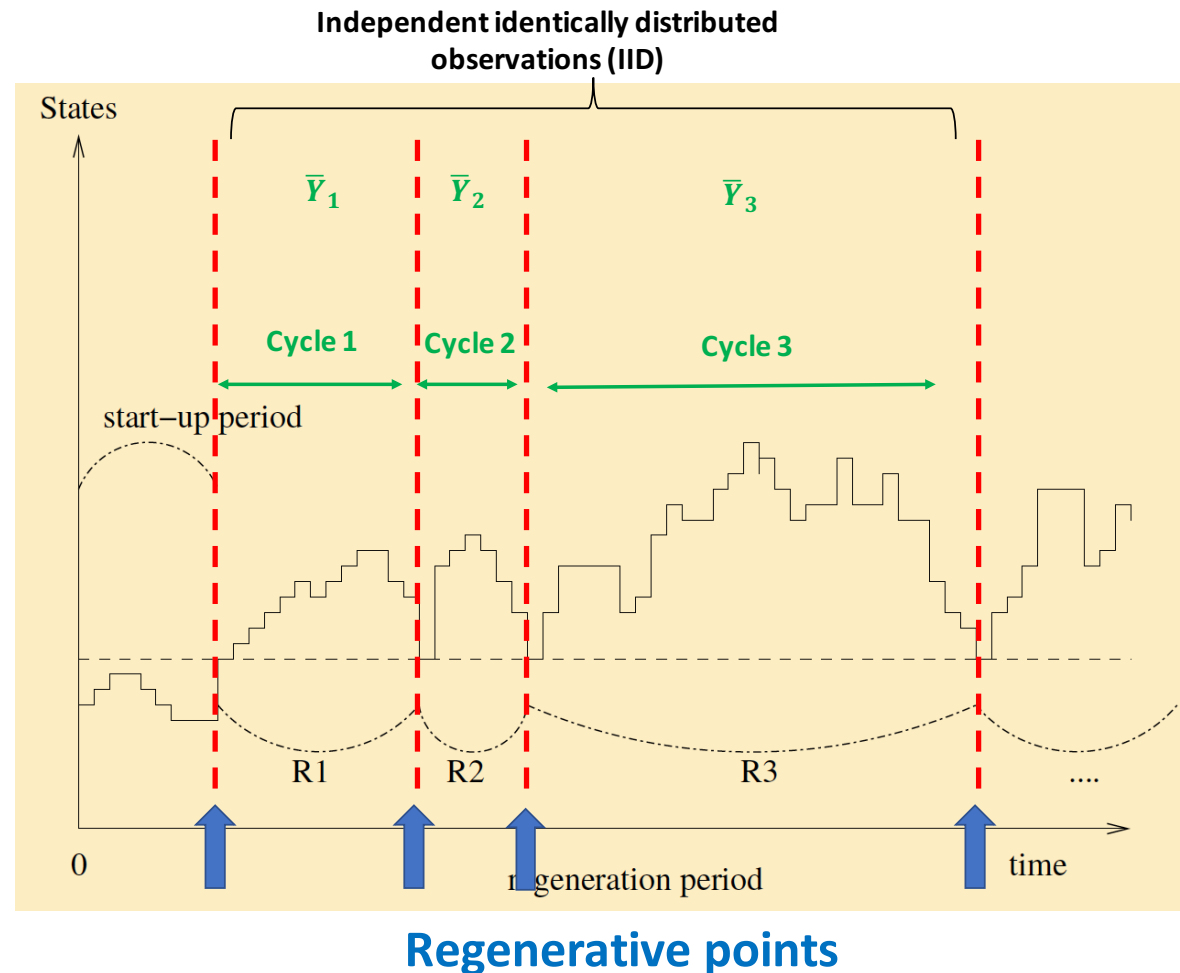
- Single run of the Markov Chain/Discrete Event Simulation Model
 - The regeneration method divides the samples in batches similar to the batch means method but now the batches start whenever the Markov Chain is in a particular state
 - Method reduces the problem of correlation between batches of Batch Mean method
 - The method is applicable to regenerative systems, i.e. systems with at least one regenerative state
 - The stochastic development of the system from that point on does not at all depend on how the state has been reached
 - Every state of a Markovian system is regenerative (e.g. M/M/c queue)
 - In a G/G/1 queue the state where the system is empty is a regenerative state
 - When the system is in the regenerative state, the system regenerates itself, i.e. the system starts 'a new life' which does not depend on the past



Estimator for Markov Chains

- Estimation Method: Regeneration method

- Principle



Estimator for Markov Chains

- Estimation Method: Regeneration method

- Principle

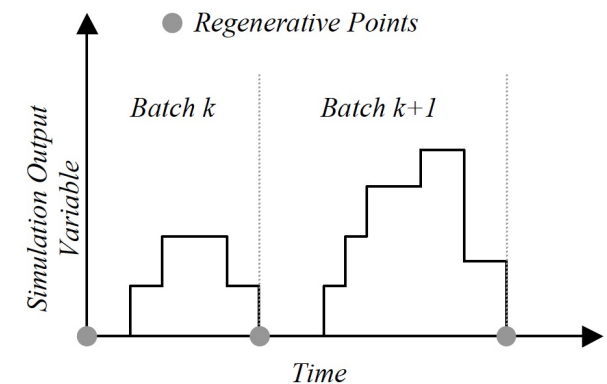
- The time instant when the system returns to the regenerative state is called the regeneration time point
 - The period between two regeneration points is called the regeneration period
 - Let $\tau_n^{(\ell)}$ be the n th time where the Markov Chain is in state ℓ

$$\tau_n^{(\ell)} = \min\{k \in \mathbb{N}, \sum_{i=0}^k \mathbb{1}\{Y_i = \ell\} = n\} \quad (\text{regenerative state – e.g. zero value})$$

and let $R_n^{(\ell)}$ be the time between the n th and the $n+1$ st visit to state ℓ

$$R_n^{(\ell)} = \tau_{n+1}^{(\ell)} - \tau_n^{(\ell)} \quad (\text{regenerative time points})$$

- Disadvantages
 - Recognize regenerative state (select zero value)
 - The time interval between regenerative batches may be (too) long



Estimator for Markov Chains

- Estimation Method: Regeneration method

- Estimators

- The mean is estimated as follows

$$\bar{X} = E[X] = \frac{\bar{Y}}{\bar{R}} \quad \text{with} \quad \bar{Y} = \frac{1}{\tilde{N}} \sum_{n=1}^{\tilde{N}} \tilde{Y}_n^{(\ell)} \quad \text{and} \quad \bar{R} = \frac{1}{\tilde{N}} \sum_{n=1}^{\tilde{N}} R_n^{(\ell)}$$

$$\bar{X} = \frac{\sum_{n=1}^{\tilde{N}} \tilde{Y}_n^{(\ell)}}{\sum_{n=1}^{\tilde{N}} R_n^{(\ell)}} \quad \text{with} \quad \tilde{Y}_n^{(\ell)} = \sum_{k=\tau_n^{(\ell)}}^{\tau_{n+1}^{(\ell)}-1} f(Y_k)$$

$\tilde{Y}_n^{(\ell)}$ a sequence of independent and identically distributed random variables and is a consistent estimator

$\tilde{N}$ the number of identified regenerative cycles

Estimator for Markov Chains

- Estimation Method: Regeneration method

- Estimators

- The variance of the sample mean can be estimated by

$$\text{var}[\bar{X}] = \underbrace{\frac{1}{\tilde{N}-1} \sum_{n=1}^{\tilde{N}} (\tilde{Y}_n - \bar{Y})^2}_{\text{Sample variance Y}} + \underbrace{(\bar{X})^2 \frac{1}{\tilde{N}-1} \sum_{n=1}^{\tilde{N}} (R_n - \bar{R})^2}_{\text{Sample variance R}} - \underbrace{2\bar{X} \frac{1}{\tilde{N}-1} \sum_{n=1}^{\tilde{N}} (\tilde{Y}_n - \bar{Y})(R_n - \bar{R})}_{\text{Covariance Y and R}}$$

- Confidence interval

$$\Pr \left[\bar{X} - k_p \frac{\sqrt{\text{var}[\bar{X}]}}{\bar{R}\sqrt{n}} \leq \mu \leq \bar{X} + k_p \frac{\sqrt{\text{var}[\bar{X}]}}{\bar{R}\sqrt{n}} \right] = p$$

Estimator for Markov Chains

- Estimation Method: Regeneration method

- Remarks

- This estimator does not delete the initial sequence of samples corresponding to the transient period
 - Estimation is started in a renewal point (and thus even the first measurements are used in estimating the performance) => The estimator is biased
 - The data of the first incomplete cycle is ignored
 - Not every process possesses regeneration points (see Glynn, P.W., and Iglehart, D.L., 1993, Conditions for the Applicability of the regenerative method, Management Science, 39, 1108-1111)

Estimator for Markov Chains

- Estimation of output parameters
 - Type of output parameters:
 - Batch means method and regenerative method are especially for output parameters that concern a single run, e.g.
 - The total time the system has spent in a blocking state
 - The time needed to process 1000 tasks
 - The total number of items put in inventory
 - The maximum number of customers in steady state in the system

Estimator for Markov Chains

- Example: The Galton-Watson process with immigration
 - Branching stochastic process to model the extinction of family names
 - Apart from offspring generation, there is also an influx of newcomers
 - Assumptions
 - Family names are passed from father to son
 - The number of different men's sons are independent random variables all having the same distribution
 - Model

$$Y_{k+1} = \sum_{\ell=1}^{Y_k} X_{k,\ell} + B_k$$

with Y_k the number of individuals in the k^{th} generation

$X_{k,\ell}$ the number of offspring of the ℓ^{th} individual in the k^{th} generation

B_k the number of offspring as a result of immigration in the k^{th} generation

Estimator for Markov Chains

- Example: The Galton-Watson process with immigration
 - The offspring is Poisson distributed with mean $\lambda = 0.9$
 - The immigration is Poisson distributed with mean $\alpha = 0.4$
 - Number of generations $K = 20$
 - Initial offspring $s_0 = 5$

Estimator for Markov Chains

- Example: The Galton-Watson process with immigration

➤ Pseudocode

Initialise parameters ()

```
Procedure(){  
  for k in range (1,K)  
    printf(k, y, sum/k);  
    y = phi(y);  
    sum += y;}
```

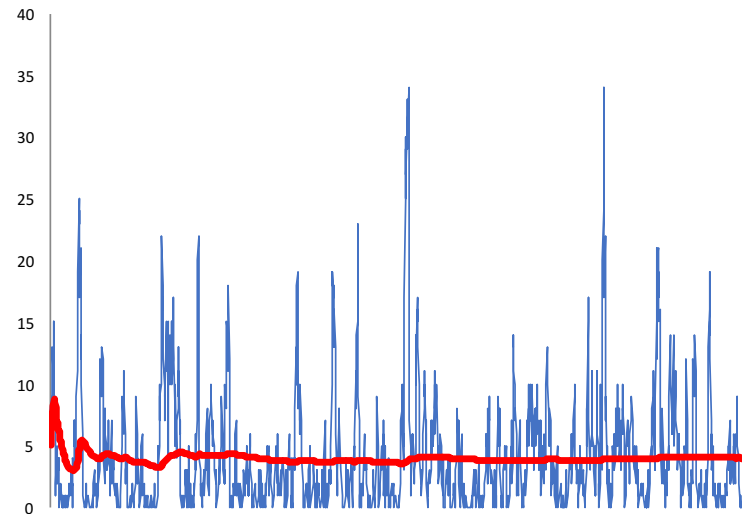
```
phi(y){  
  offspring = 0  
  for i in range (1, y)  
    offspring += Poisson_distribution(lambda);  
  offspring += Poisson_distribution(alfa);  
  return offspring;}
```


Estimator for Markov Chains

- Example: The Galton-Watson process with immigration

➤ Solution – 1 run (n = 2000 generations)

- Markov trajectory: Blue line
- Running average: Red line



- Average number of offspring $\bar{X} = 3.96$

Estimator for Markov Chains

- Example: The Galton-Watson process with immigration

- Solution – 10 simulation runs

- Evaluation of the long-run average

	Average
Run 1	3.9605
Run 2	3.9325
Run 3	3.8675
Run 4	4.1715
Run 5	3.7215
Run 6	4.248
Run 7	3.582
Run 8	3.7565
Run 9	4.0855
Run 10	3.7105
Avg	3.9036
St dev	0.21714266
CI-	3.72705833
CI+	4.08014167

Beware of the transient period

⇒ There may be a bias in the results

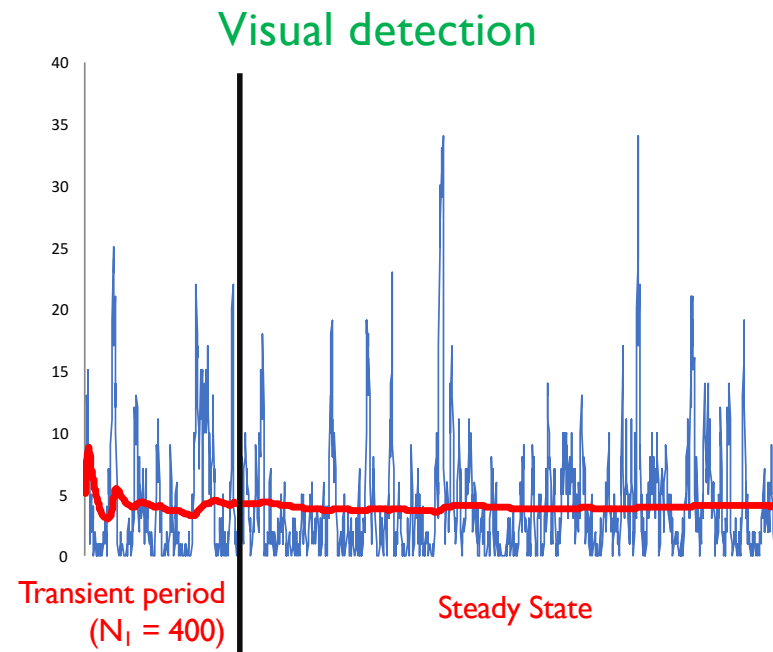
⇒ Delete the first N_i observations of each run

95% confidence interval – Use student-t distribution ($n = 10$)

Estimator for Markov Chains

- Example: The Galton-Watson process with immigration
 - Solution – 10 simulation runs
 - Evaluation of the long-run average without transient period

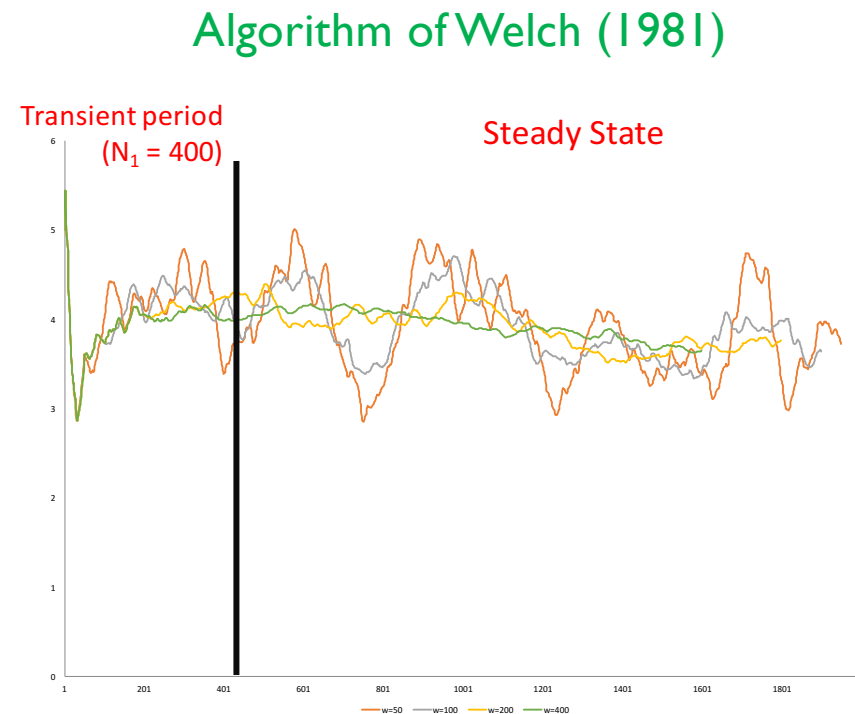
	Average
Run 1	3.91
Run 2	4.49
Run 3	4.24
Run 4	4.06
Run 5	3.58
Run 6	4.46
Run 7	3.84
Run 8	4.01
Run 9	3.75
Run 10	3.70
Avg	4.00
St dev	0.31
CI-	3.75
CI+	4.26



Estimator for Markov Chains

- Example: The Galton-Watson process with immigration
 - Solution – 10 simulation runs
 - Evaluation of the long-run average without transient period

	Average
Run 1	3.91
Run 2	4.49
Run 3	4.24
Run 4	4.06
Run 5	3.58
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Run 8	4.01
Run 9	3.75
Run 10	3.70
Avg	4.00
St dev	0.31
CI-	3.75
CI+	4.26



Estimator for Markov Chains

- How to evaluate transient behaviour?
 - For Markov Chains that do not reach steady-state or output measures that are not related to steady state
 - The estimator $\bar{X}$ is a consistent estimate of the transient behaviour for transient Markov Chains

$$\bar{X} = \frac{1}{N} \sum_{n=1}^N f(Y_n)$$

- Determine natural event that specifies the length of each run
- Methods
 - Replication method: Multiple runs
 - The different runs use independent seeds and the same initialisation rule
 - The initial conditions affect the desired measure of performance and should be representative for the system

Estimator for Markov Chains

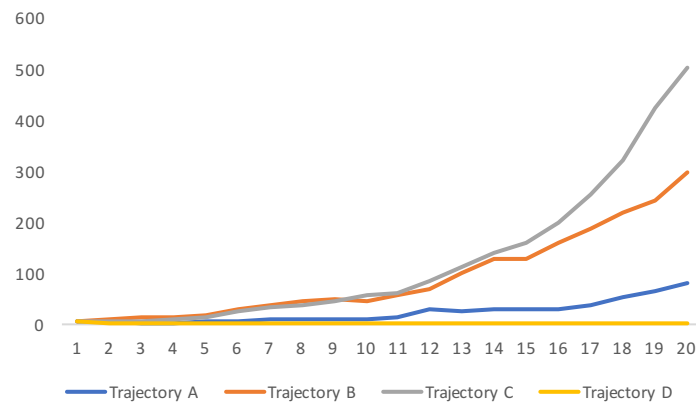
- Example – The Galton-Watson process without immigration
 - Question: Calculate the average number of persons with the same family name over 20 generations ($K = 20$, $s_0 = 5$, $\lambda = 1.2$)
 - Markov Chain
 - Within-replication data are dependent
 - Across-replication data are independent => Multiple Markov trajectories need to be evaluated
 - Markov Chain is not ergodic
 - Terminating simulation with fixed stop criterion = 100 runs
 - => Use replication method with different seeds
 - => The estimator $\bar{X}$ is a consistent estimate of the transient behaviour for transient Markov Chains

$$\bar{X} = \frac{1}{N} \sum_{n=1}^N f(Y_n)$$

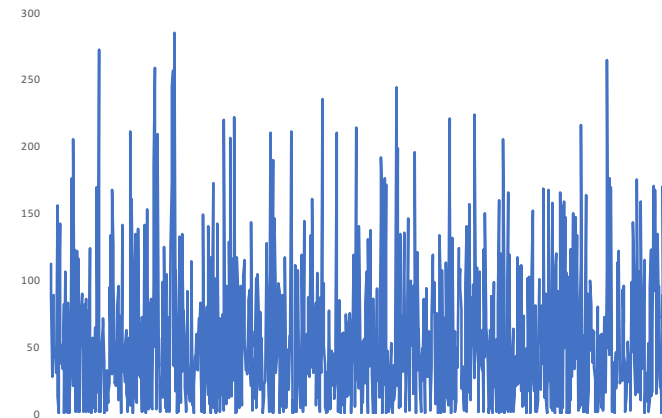
Estimator for Markov Chains

- Example – The Galton-Watson process without immigration
 - Question: Calculate the average number of persons with the same family name over 20 generations ($K = 20$, $s_0 = 5$, $\lambda = 1.2$)
 - Simulation of $n = 1000$ experiments consisting of a single run to determine the running average
 - Graphs

Different example Markov Trajectories

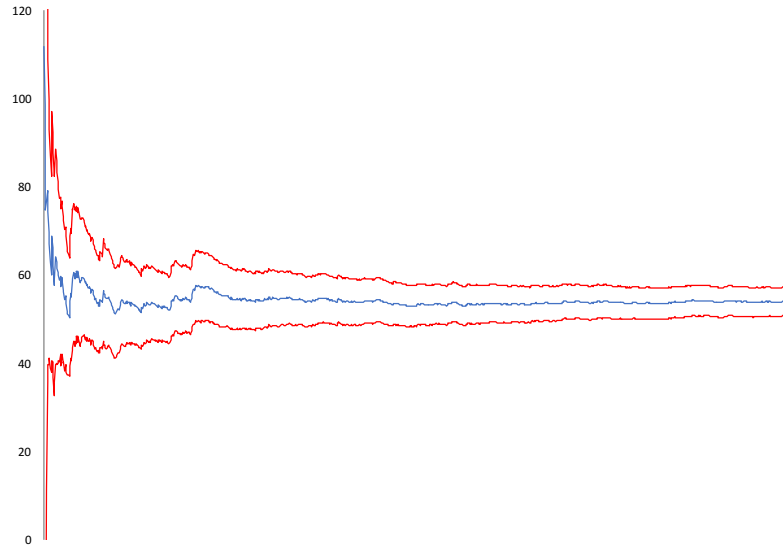


Average per Markov Trajectory



Estimator for Markov Chains

- Example – The Galton-Watson process without immigration
 - Question: Calculate the average number of persons with the same family name over 20 generations ($K = 20$, $s_0 = 5$, $\lambda = 1.2$)
 - Simulation of $n = 1000$ experiments consisting of a single run to determine the running average
 - Point estimator: $\bar{X} = 54.24$
 - Interval estimator: $[51.05, 57.44]$
 - Graph
 - Running average (blue line)
 - Confidence interval (red line)



Variance reduction for Markov Chains

- Goal variance reduction
 - The standard deviation of a Monte Carlo estimate is always equal to a constant divided by the square root of the number of samples
 - The accuracy may be improved by
 - Increasing the number of samples
 - Decreasing the variance
 - Techniques to reduce the variance (cf. Previous topic)
 - Antithetic variables
 - Control variables
 - Etc

Variance reduction for Markov Chains

- Antithetic variables

- Principle

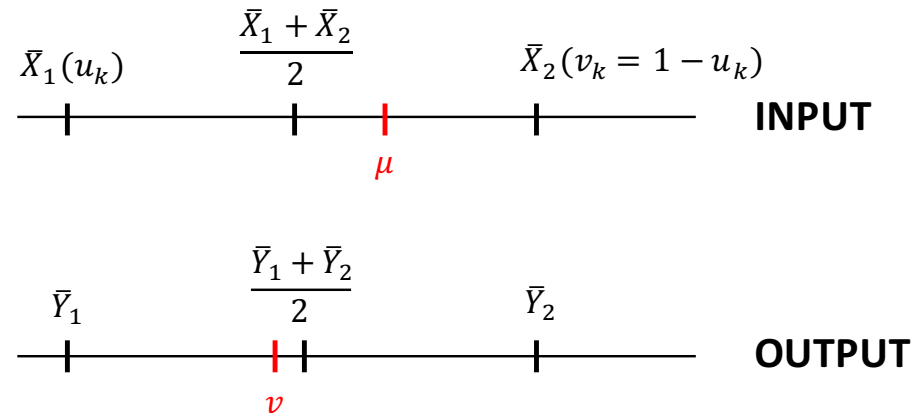
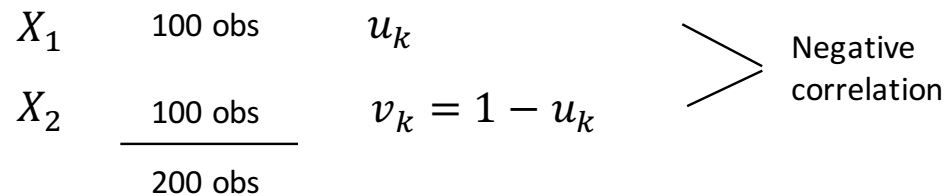
- The antithetic method attempts to introduce negative correlation into the samples which decreases the variance
 - An overestimate is paired as much as possible with an underestimate
 - Taking the average of such an antithetic pair (an overestimate and an underestimate) gives a sample with a considerably reduced standard deviation
 - Antithetic variates can be applied as soon as one can show that samples to be generated are monotonically dependent (either monotonically increasing or monotonically decreasing) on certain random numbers that one generates.

- Remark

- Very easy to implement
 - The reduction of the standard deviation is not spectacular

Variance reduction methods

- Antithetic variables or complementary random numbers



Variance reduction for Markov Chains

- Antithetic variables

- Example: Waiting times in a queue

- Lindley equation: The consecutive waiting times relate as

$$Y_{k+1} = \max(Y_k + S_k - A_k, 0)$$

=> Delay times are monotonically decreasing functions of the interarrival times

- Trajectory 1: Longer interarrival times before item k arrives, will lead to a smaller delay D_k^1 for item k
- Trajectory 2: Smaller interarrival times generated before item k arrives, will lead to a longer delay D_k^2 for item k

=> The average of both trajectories has a much smaller variance compared to a single trajectory

=> Delay times are monotonically increasing functions of the service times

- The delay of the k^{th} increases whenever the service time of this item and/or that of its predecessors become larger

The monotonicity assumption to ensure a negative correlation is satisfied

Sample paths can be calculated for the original variables and the antithetic variables

Variance reduction for Markov Chains

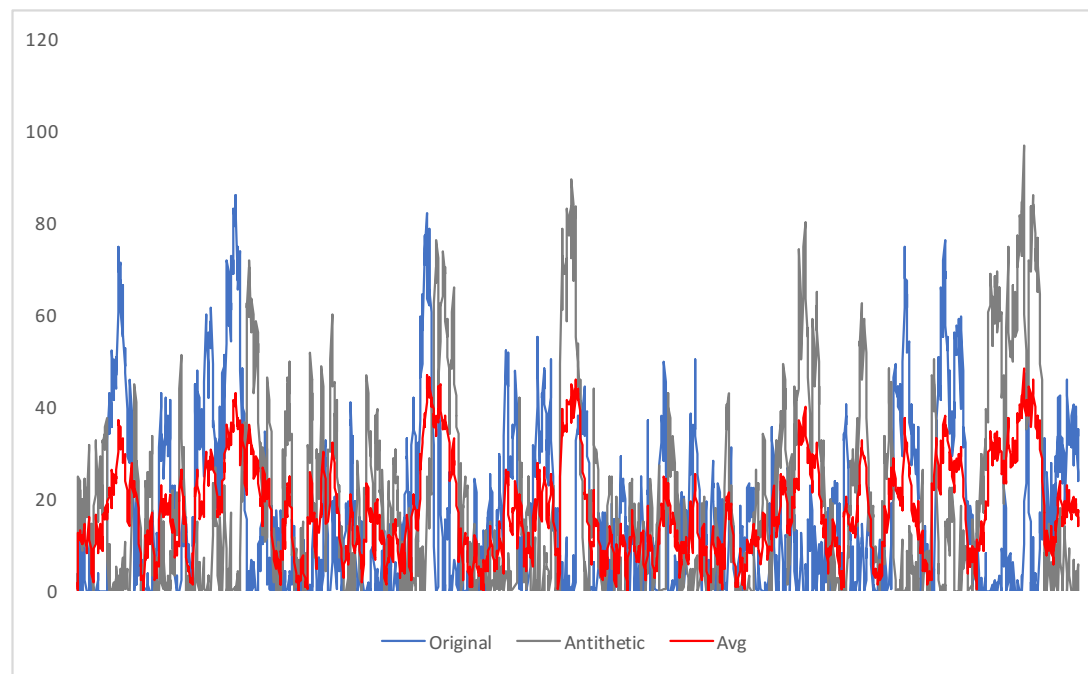
- Antithetic variables
 - Example: Waiting times in a queue
 - Estimators

$$\bar{X}^{anti} = \frac{1}{N} \sum_{k=1}^N \frac{D_k^1 + D_k^2}{2} \quad \text{and} \quad \text{var}[\bar{X}^{anti}] = \frac{1}{4N} (\text{var}[D_k^1] + \text{var}[D_k^2] + 2\text{cov}[D_k^1 D_k^2])$$

For complicated or large systems, there are too many other random variables influencing the delay of the items. The correlation of the delay with interarrival and service times becomes small, which diminishes the variance reduction that can be achieved.

Variance reduction for Markov Chains

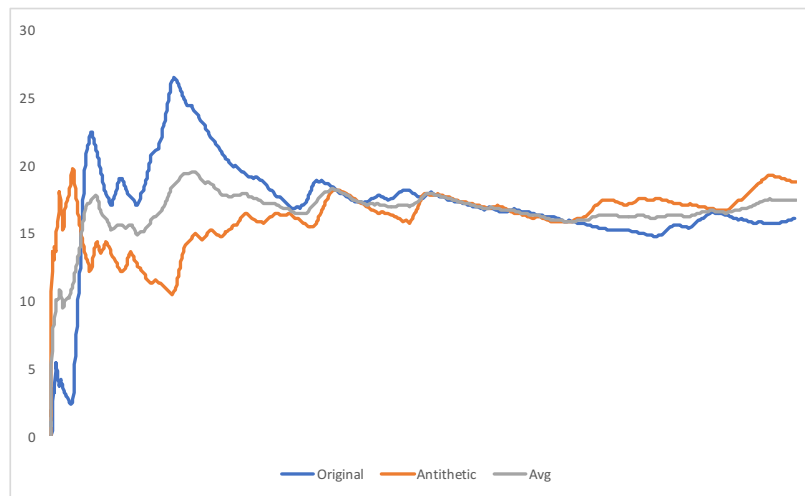
- Antithetic variables
 - Example: Waiting times in a queue
 - Results: State variable – Waiting time



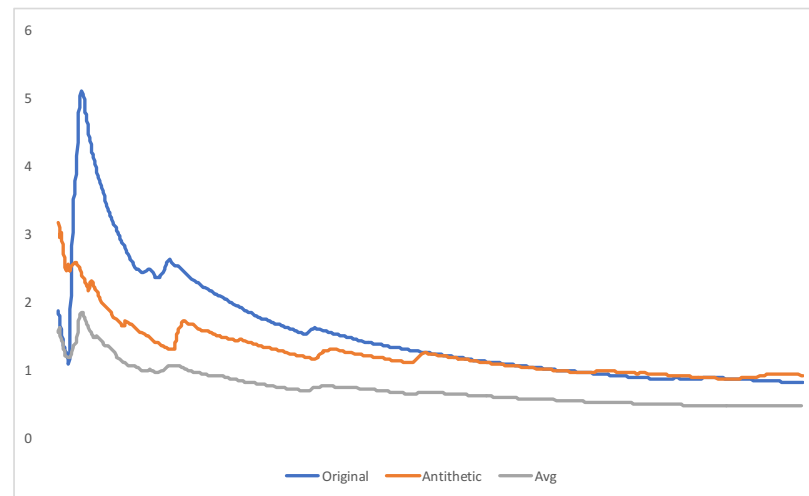
Variance reduction for Markov Chains

- Antithetic variables
 - Example: Waiting times in a queue
 - Results: Running average – Waiting time

Running average



Accuracy (half-width)



Variance reduction for Markov Chains

- Control variables

- Principle

- Monotone relationship between original variable and control variable, i.e. $X_k = wY_k$
 - Y_k has some predictive power for X_k (correlation)
 - Suppose $E[Y_k] = v$ and can be calculated analytically
 - Use $Y_k - E[Y_k]$ as a control variable to mitigate the samples X_k , i.e.

$$\bar{X}_c = \frac{1}{n} \sum_{k=1}^n X_k - c(Y_k - E[Y_k])$$

- If $\bar{X}$ is an unbiased estimator of X , $\bar{X}_c = \bar{X} - c(\bar{Y} - v)$ is also an unbiased estimator
 - For estimation of $\bar{X}_c$ of a complicated system, use a trajectory of X_k of a related simple system (e.g. M/M/1 queue) as control, for which the performance is known analytically

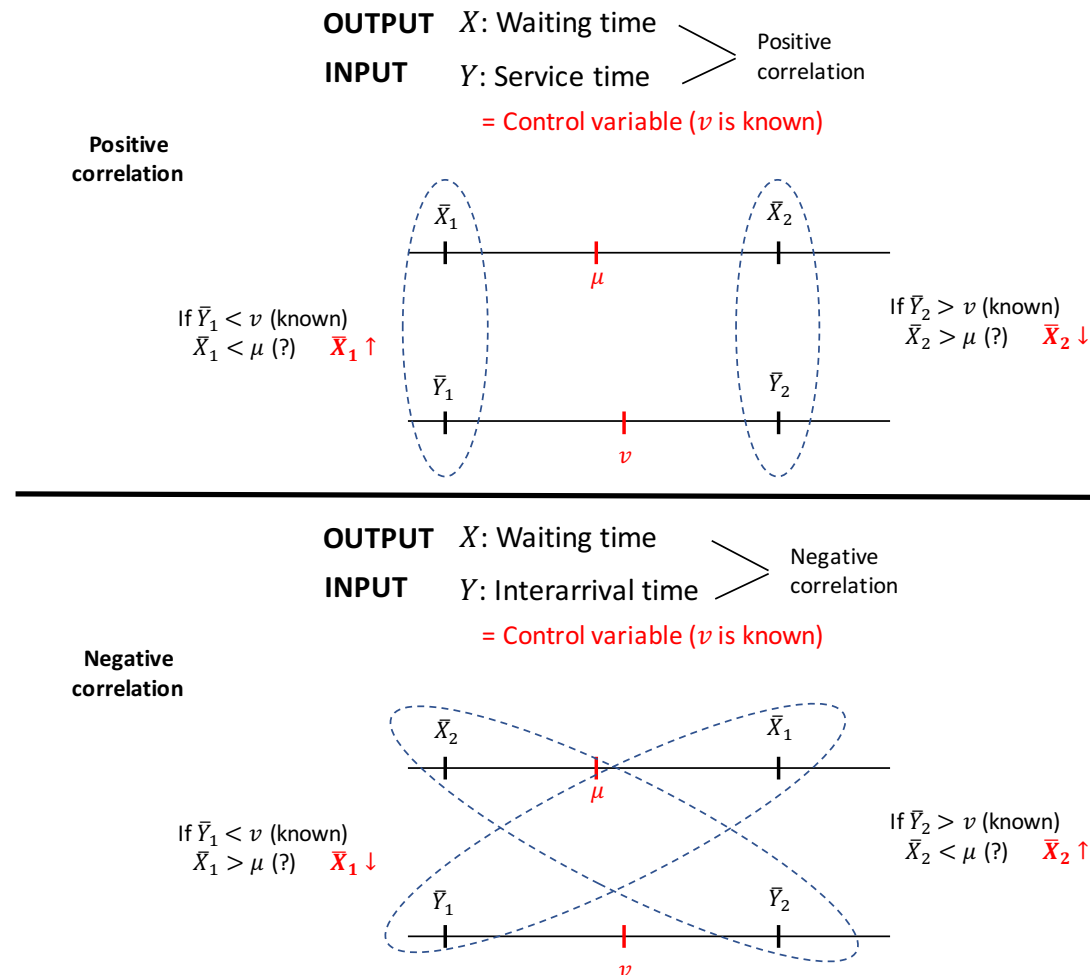
=> Important to ensure a strong correlation

Variance reduction methods

- Control variates

- The principle

(not in your slides)



Variance reduction for Markov Chains

- Control variables

- Example: Waiting times in a queue

- Since interarrival times τ_k and service times S_k (with known mean value) are usually monotonically related to delay times, we can make use of them as control variables as follows

- $\bar{D}_{k,c} = \bar{D}_k + c_A(\bar{\tau}_k - E[\tau_k])$ with $E[\tau_k] = 1/\lambda$

- $\bar{D}_{k,c} = \bar{D}_k + c_S(\bar{S}_k - E[S_k])$ with $E[S_k] = 1/\mu$

=> Both of them will lead to a variance reduction with $c_A > 0$ and $c_S < 0$

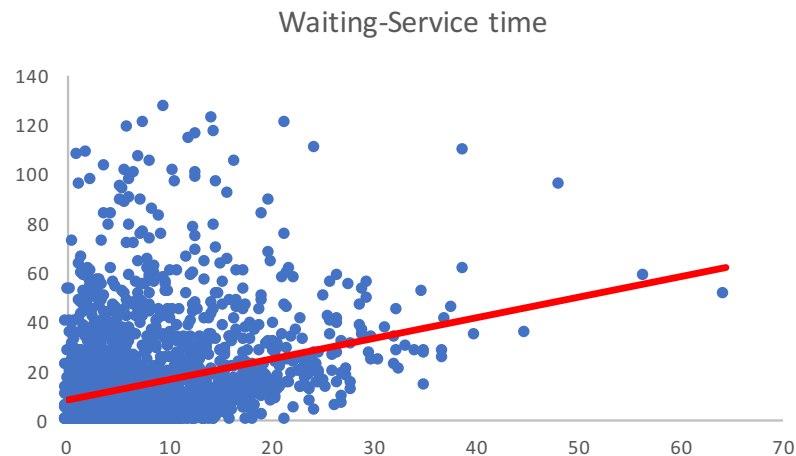
- Remark

- For complicated or large systems, there are too many other random variables influencing the delay of the items. The correlation of the delay with interarrival and service times becomes small, which diminishes the variance reduction that can be achieved.

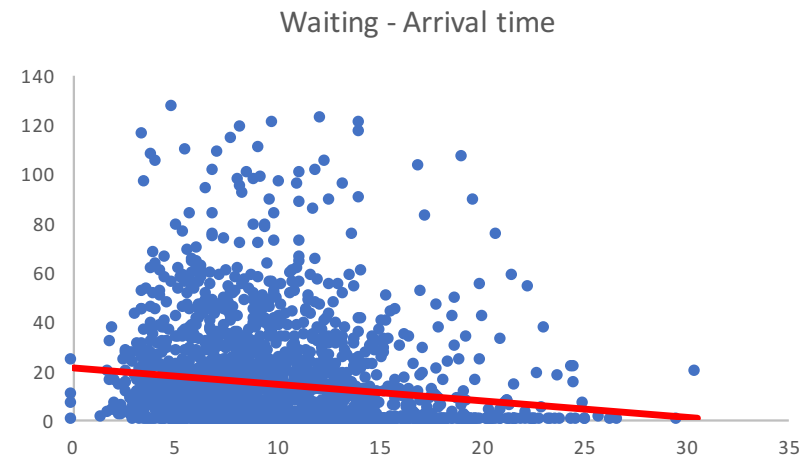
Variance reduction for Markov Chains

- Control variables
 - Example: Waiting times in a queue

Waiting – Service time



Waiting – Arrival time



Variance reduction for Markov Chains

- Control variables

- Example: Waiting times in a queue

- $\bar{X}_c = \bar{X} - c(\bar{Y} - v)$
- $\text{Var}[\bar{X}_c] = \text{Var}[\bar{X}] + c^2\text{Var}[\bar{Y}] - 2c\text{Cov}(\bar{X}, \bar{Y})$

	Original (X)	Control variables (Y)		
	Waiting time	Service time (1)	Arrival time (2)	(1)+(2)
Sample mean	14.69	8.02	9.99	
Mean (μ)		8	10	
Var (S)	413.29	49.55	20.20	
Cov (X,Y)		41.45	-13.43	
c_{opt}		0.84	-0.66	
$\bar{X}_c$		14.67	14.68	14.66
$\text{Var}[\bar{X}_c]$	413.29	378.61	404.36	369.68
n	2000	2000	2000	2000
Halfwidth (95%)	0.891	0.853	0.881	0.843